

Abstract Algebra I

Turja Roy

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0 Sets and Relations

0.1 Sets

Definition 0.1.1: Set

A **set** is a well-defined collection of distinct objects, considered as an object in its own right. The objects in a set are called **elements** or **members** of the set.

1. A set is made up of **elements**, and if a is one of these elements, we denote this fact by $a \in S$.
2. There is exactly one set with no elements. It is the **empty set** and is denoted by $\{\}$ or $\emptyset$.
3. We may describe a set either by giving a characterizing property of the elements, such as $S = \{x \mid P(x)\}$, or by listing the elements, such as $S = \{a, b, c\}$.
4. A set is **well-defined**, meaning that if S is a set and a is an object, then either a is definitely in S , denoted by $a \in S$, or a is definitely not in S , denoted by $a \notin S$.

Definition 0.1.2: Subset

A set A is a **subset** of a set B , denoted $A \subseteq B$, or $B \supseteq A$, if every element of A is also an element of B .

Definition 0.1.3: Proper and Improper Subsets

A **proper subset** of a set B is a subset A such that $A \neq B$, and is denoted by $A \subset B$ or $B \supset A$. An **improper subset** of a set B is the set B itself, denoted by $B \subseteq B$.

Definition 0.1.4: Cartesian Product

The **Cartesian product** of two sets A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$. Formally,

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

0.2 Relations

Definition 0.2.1: Relation

A **relation** between sets A and B is a subset $\mathcal{R}$ of $A \times B$. We read $(a, b) \in \mathcal{R}$ as "a is related to b" and write $a\mathcal{R}b$.

Relation on S

Any relation between a set S and itself is called a **relation on S** .

Definition 0.2.2: Mapping

A **function** ϕ mapping X into Y is a relation between X and Y with the property that each $x \in X$ appears as the first member of exactly one ordered pair (x, y) in ϕ . Such a function is called a **map** or **mapping** of X into Y . We write $\phi : X \rightarrow Y$ and express $(x, y) \in \phi$ by $\phi(x) = y$.

Domain, Codomain, Range

For the mapping $\phi : X \rightarrow Y$,

- The **domain** of ϕ is the set X
- The set Y is the **codomain** of ϕ
- The **range** of ϕ is $\phi[X] = \{\phi(x) \mid x \in X\}$.

0.3 Cardinality

Definition 0.3.1: Cardinality

The number of elements in a set S is called the **cardinality** of S , denoted by $|S|$. If S is finite, then $|S|$ is a non-negative integer. If S is infinite, then $|S|$ is not a finite number.

Definition 0.3.2: Injective, Surjective, and Bijective Functions

A function $\phi : A \rightarrow B$ is **injective** (or **one-to-one**) if

$$\forall a_1, a_2 \in A, \quad \phi(a_1) = \phi(a_2) \implies a_1 = a_2.$$

It is **surjective** (or **onto**) if the range of ϕ is B , i.e.,

$$\forall b \in B, \exists a \in A \text{ s.t. } \phi(a) = b$$

It is **bijective** if it is both injective and surjective.

Definition 0.3.3: One-to-one Correspondence

A **one-to-one correspondence** between two sets A and B is a bijective function $\phi : A \rightarrow B$. If such a function exists, we say that A and B have the same cardinality, denoted by $|A| = |B|$.

Definition 0.3.4: Equal Cardinality

Two sets A and B have the **same/equal cardinality** if there exists a one-to-one function mapping A onto B , that is, if there exists a one-to-one correspondence between A and B . In this case, we write $|A| = |B|$.

Example 0.3.1

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is not injective, since $f(2) = f(-2) = 4$ but $2 \neq -2$. Also it is not surjective, since there is no $x \in \mathbb{R}$ such that $f(x) = -1$. However, the function $g : \mathbb{R} \rightarrow \mathbb{R}$ where $g(x) = x^3$ is both injective and surjective, since for any $y \in \mathbb{R}$, there exists a unique $x \in \mathbb{R}$ such that $g(x) = y$, namely $x = \sqrt[3]{y}$. Therefore, g is a bijection.

Example 0.3.2: Show that $\mathbb{Z}$ and $\mathbb{Z}^+$ have the same cardinality

We can define a function $f : \mathbb{Z} \rightarrow \mathbb{Z}^+$ by

$$f(n) = \begin{cases} 2n + 1 & \text{if } n \geq 0, \\ -2n & \text{if } n < 0. \end{cases}$$

This function is injective, since if $f(n_1) = f(n_2)$, then either both n_1 and n_2 are non-negative or both are negative. In the first case, we have $2n_1 + 1 = 2n_2 + 1$, which implies $n_1 = n_2$. In the second case, we have $-2n_1 = -2n_2$, which also implies $n_1 = n_2$. Therefore, f is injective.

To show that f is surjective, let $m \in \mathbb{Z}^+$. If m is odd, then there exists a non-negative integer k such that $m = 2k + 1$, and we can choose $n = k$ to get $f(n) = m$. If m is even, then there exists a positive integer k such that $m = 2k$, and we can choose $n = -k$ to get $f(n) = m$. Therefore, for every positive integer m , there exists an integer n such that $f(n) = m$, which means that f is surjective.

Since f is both injective and surjective, it is a bijection. Therefore, $\mathbb{Z}$ and $\mathbb{Z}^+$ have the same cardinality.

Note:-

It is fascinating that a proper subset of an infinite set may have the same number of elements as the whole set; an **infinite set** can be defined as a set having this property.

0.4 Partitions and Equivalence Relations

Definition 0.4.1: Partition

A **partition** of a set S is a collection of nonempty subsets of S such that every element of S is in exactly one of the subsets. The subsets are the **cells** of the partition.

Example 0.4.1

Splitting $\mathbb{Z}$ into the subset of even integers and the subset of odd integers, we obtain a partition of $\mathbb{Z}$ into the two cells listed below:

$$\begin{aligned} [0] &= \{\dots, -8, -6, -4, -2, 0, 2, 4, \dots\} \\ [1] &= \{\dots, -7, -5, -3, -1, 1, 3, 5, \dots\} \end{aligned}$$

We can think of $[0]$ as being the integers that are divisible by 2, and $[1]$ as the integers when divided by 2 yield a remainder of 1. This idea can be used for positive integers other than 2 as well. For example, we can partition $\mathbb{Z}$ into three cells:

$$\begin{aligned} [0] &= \{x \in \mathbb{Z} \mid x \equiv 0 \pmod{3}\}, \\ [1] &= \{x \in \mathbb{Z} \mid x \equiv 1 \pmod{3}\}, \\ [2] &= \{x \in \mathbb{Z} \mid x \equiv 2 \pmod{3}\}. \end{aligned}$$

Definition 0.4.2: Equivalence Relation

An **equivalence relation** $\mathcal{R}$ on a set S is one that satisfies these three properties for all $x, y, z \in S$:

1. **Reflexivity:** $x\mathcal{R}x$.
2. **Symmetry:** If $x\mathcal{R}y$, then $y\mathcal{R}x$.
3. **Transitivity:** If $x\mathcal{R}y$ and $y\mathcal{R}z$, then $x\mathcal{R}z$.

Example 0.4.2

For any nonempty set S , the equality relation $=$ defined by the subset $\{(x, x) \mid x \in S\}$ of $S \times S$ is an equivalence relation on S . This relation is reflexive, since every element x of S is related to itself. It is symmetric, since if x and y are related, then y and x are also related. It is transitive, since if x and y are related and y and z are related, then x and z are also related. Therefore, the equality relation is an equivalence relation on any nonempty set.

Another example of an equivalence relation on $\mathbb{Z}$ is the relation $\mathcal{R}$ defined by $x\mathcal{R}y$ if and only if x and y have the same parity, that is, both are even or both are odd. This relation is reflexive, since every integer has the same parity as itself. It is symmetric, since if x and y have the same parity, then y and x also have the same parity. It is transitive, since if x and y have the same parity and y and z have the same parity, then x and z also have the same parity. Therefore, $\mathcal{R}$ is an equivalence relation on $\mathbb{Z}$.

Definition 0.4.3: Equivalence Class

If $\mathcal{R}$ is an equivalence relation on a set S and $x \in S$, then the **equivalence class** of x under $\mathcal{R}$ is the set of all elements in S that are equivalent to x under $\mathcal{R}$, denoted by $[x]$ or $\bar{x}$. Formally,

$$[x] \text{ (or } \bar{x}) = \{y \in S \mid y\mathcal{R}x\}.$$

In example 0.4.1, the equivalence classes under the relation $\mathcal{R}$ are $[0]$ and $[1]$, which consist of the even integers and the odd integers, respectively. Here, the even and odd integers are equivalent to 0 and 1 respectively under the modulo 2 operation in a sense that they yield the same remainder when divided by 2.

Definition 0.4.4: Congruence Modulo n

For a positive integer n , the relation $\equiv_n$ on $\mathbb{Z}$ defined by $x \equiv_n y$ if and only if x and y have the same remainder when divided by n is an equivalence relation on $\mathbb{Z}$. This relation is called **congruence modulo n** .

Another way of writing is:

$$x \equiv_n y \iff x \equiv y \pmod{n} \iff n \mid (x - y).$$

Theorem 0.4.1 (Equivalence Relations and Partitions): Let S be a nonempty set and let $\sim$ be an equivalence relation on S . Then $\sim$ yields a partition of S , where

$$[a] = \{x \in S \mid x \sim a\}$$

Also, each partition of S gives rise to an equivalence relation $\sim$ on S where $a \sim b$ if and only if a and b are in the same cell of the partition.

Proof:

Let $\sim$ be an equivalence relation on S . We will show that the collection of equivalence classes $\{[a] \mid a \in S\}$ forms a partition of S .

First, we need to show that each equivalence class $[a]$ is nonempty. Since $\sim$ is reflexive, we have $a \sim a$, which means that $a \in [a]$. Therefore, $[a]$ is nonempty.

Next, we need to show that the equivalence classes are disjoint. Suppose that $[a] \cap [b] \neq \emptyset$ for some $a, b \in S$. Then there exists an element x such that $x \in [a]$ and $x \in [b]$. This means that $x \sim a$ and $x \sim b$. By symmetry and transitivity of $\sim$, we have $a \sim x$ and $x \sim b$, which implies that $a \sim b$. Therefore, if two equivalence classes intersect, they must be the same class. Hence, the equivalence classes are disjoint.

Finally, we need to show that the union of all equivalence classes is equal to S . Let x be an arbitrary element of S . Since $\sim$ is reflexive, we have $x \sim x$, which means that $x \in [x]$. Therefore, every element of S belongs to at least one equivalence class. Hence, the union of all equivalence classes is equal to S . $\square$

Lemma 0.4.2: Each cell in the partition of S induced by an equivalence relation $\sim$ is an equivalence class under $\sim$.

1 Groups and Subgroups

1.1 Binary Operations

Definition 1.1.1: Binary Operation

A **binary operation** $*$ on a set S is a function mapping $S \times S$ into S . For each $(a, b) \in S \times S$, we will denote the element $*(a, b)$ of S by $a * b$.

Note:-

Binary refers to the fact that we are mapping *pairs* of elements from S into S . We could also define ternary operation as a function mapping triplets of elements of S to S , but that is not relevant for our discussion of groups.

Example 1.1.1

Addition $(+)$ is a binary operator on the set $\mathbb{R}$. Multiplication $(\cdot)$ is also a binary operator on $\mathbb{R}$. However, division $(/)$ is not a binary operator on $\mathbb{R}$ since a/b is not defined when $b = 0$. But, division is a binary operator on the set $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$.

Note that we require an operation on a set S to be defined for *every* ordered pair (a, b) of elements from S .

Definition 1.1.2: Closure and Induced Operation

Let $*$ be an operation on S and let H be a subset of S . The subset H is **closed under** $*$ if

$$\forall a, b \in H, a * b \in H$$

In this case, the operation on H given by restricting $*$ to H is the **operation induced** of $*$ on H .

Note:-

If $*$ is an operation on S and H is a subset of S , then we can define an operation on H by using the same rule as $*$, but only applying it to elements of H . For this to work, we need to make sure that when we apply the operation to any two elements of H , we get another element of H . If this is the case, then we say that H is closed under $*$, and the operation we defined on H is called the induced operation.

Example 1.1.2

The addition $+$ operation on $\mathbb{R}$ does not induce an operation on the set $\mathbb{R}^*$ of nonzero real numbers because $-2, 2 \in \mathbb{R}^*$, but $2 + (-2) = 0 \notin \mathbb{R}^*$. Thus, $\mathbb{R}^*$ is not closed under $+$, and so $+$ does not induce an operation on $\mathbb{R}^*$.

However, the multiplication $\cdot$ operation on $\mathbb{R}$ does induce an operation on $\mathbb{R}^*$ since for any $a, b \in \mathbb{R}^*$, we have $a \cdot b \in \mathbb{R}^*$. Thus, $\mathbb{R}^*$ is closed under $\cdot$.

Example 1.1.3: Let $+$ and $\cdot$ be the usual operations of addition and multiplication on the set $\mathbb{Z}$, and let $H = \{n^2 \mid n \in \mathbb{Z}^+\}$. Determine whether H is closed under $+$ and $\cdot$.

For $+$, we need to only observe that $1^2 = 1$ and $2^2 = 4$ are in H , but that $1 + 4 = 5$ and $5 \notin H$. This H is not closed under addition.

For $\cdot$, suppose that $r \in H$ and $s \in H$. Using that it means for r and s to be in H , we

see that there must be integers n and m in $\mathbb{Z}^+$ such that $r = n^2$ and $s = m^2$. Consequently, $rs = n^2m^2 = (nm)^2$. By the characterization of elements in H and the fact that $nm \in \mathbb{Z}^+$, this means that $rs \in H$, so H is closed under multiplication.

Example 1.1.4

Let F be the set of all real-valued functions f having as domain $\mathbb{R}$. For each ordered pair (f, g) of functions in F , we define for each $x \in \mathbb{R}$:

$f + g$ by $(f + g)(x) = f(x) + g(x)$	addition
$f - g$ by $(f - g)(x) = f(x) - g(x)$	subtraction
$f \cdot g$ by $(f \cdot g)(x) = f(x) \cdot g(x)$	multiplication
$f \circ g$ by $(f \circ g)(x) = f(g(x))$	composition

All four of these functions are again real valued with domain $\mathbb{R}$, so F is closed under all four operations $+$, $-$, $\cdot$, and $\circ$.

Definition 1.1.3: Commutativity and Associativity

An operation $*$ on a set S is **commutative** if and only if:

$$\forall a, b \in S, \quad a * b = b * a$$

An operation $*$ on a set S is **associative** if and only if:

$$\forall a, b, c \in S, \quad (a * b) * c = a * (b * c)$$

Theorem 1.1.1 (Associativity of Composition): Let S be a set and let f, g and h be functions mapping S into S . Then

$$f \circ (g \circ h) = (f \circ g) \circ h$$

Proof:

Let $x \in S$. Then we have:

$$\begin{aligned} (f \circ (g \circ h))(x) &= f((g \circ h)(x)) \\ &= f(g(h(x))) \\ &= (f \circ g)(h(x)) \\ &= ((f \circ g) \circ h)(x) \end{aligned}$$

□

Definition 1.1.4: Identity Element

Let $*$ be a binary operation on a set S . An element e of S is an **identity element** for $*$ if and only if:

$$\forall a \in S, \quad e * a = a * e = a$$

Theorem 1.1.2 (Uniqueness of Identity Element): A set with binary operation $*$ has at most one identity element.

Proof:

Let e and e' be identity elements for $*$. Consider the element $e * e'$. Since e is an identity element, we have $e * e' = e'$. But $e * e' = e$ since e' is an identity element. Thus, $e = e'$, so there is at most one identity element for $*$. $\square$

Definition 1.1.5: Inverse Element

If $*$ is an operation on the set S , and S has an identity element e , then for any $x \in S$, the **inverse** of x is an element x' such that

$$x * x' = x' * x = e$$

1.2 Groups**Definition 1.2.1: Group**

A **group** $\langle G, * \rangle$ is a set G , closed under a binary operation $*$, such that the following axioms are satisfied:

$\mathcal{G}_1$: (**Associativity of $*$**) For all $a, b, c \in G$, we have

$$(a * b) * c = a * (b * c)$$

$\mathcal{G}_2$: (**Identity element e for $*$**) There is an element e in G such that for all $x \in G$,

$$e * x = x * e = x$$

$\mathcal{G}_3$: (**Inverse a' of a**) For each $a \in G$, there is an element a' in G such that

$$a * a' = a' * a = e$$

Example 1.2.1

$\langle \mathbb{R}, + \rangle$ is a group with identity element 0 and the inverse of any real number a is $-a$. However, $\langle \mathbb{R}, \cdot \rangle$ is not a group since 0 has no multiplicative inverse.

Definition 1.2.2: Abelian Group

A group G is an **Abelian group** (or commutative group) if its binary operation is commutative. That is,

$$\forall a, b \in G, \quad a * b = b * a$$

Example 1.2.2

$\langle \mathbb{Z}^+, + \rangle$ is not a group since there is no identity element for $+$ in $\mathbb{Z}^+$.

The set of all nonnegative integers (including 0) is still not a group under $+$ since there is no inverse for any positive integer.

However, the sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are all groups under $+$, and they are all Abelian groups since addition is commutative.

Example 1.2.3: Linear Algebra

Axioms for a vector space V pertaining just to vector addition can be summarized by

asserting that V under vector addition is an Abelian group. The identity element is the zero vector $\vec{0}$, and the inverse of a vector $\vec{v}$ is $-\vec{v}$.

Definition 1.2.3: General Linear Group of Degree n

The group of all invertible $n \times n$ matrices under matrix multiplication is called the **general linear group of degree n** and is denoted by $GL_n(\mathbb{R})$ or $GL(n, \mathbb{R})$.

Example 1.2.4

The set $M_n(\mathbb{R})$ of all $n \times n$ matrices under matrix multiplication is not a group. The $n \times n$ matrix with all entries 0 has no inverse.

However, the set $GL_n(\mathbb{R})$ of all invertible $n \times n$ matrices under matrix multiplication is a group. The identity element is the $n \times n$ identity matrix, and the inverse of an invertible matrix A is its inverse A^{-1} .

Definition 1.2.4: Isometry of a plane

An **isometry** of a plane is a function f mapping the plane into itself such that for all points P and Q in the plane, the distance from $f(P)$ to $f(Q)$ is the same as the distance from P to Q .

Isometries of the plane map the plane one-to-one and onto itself. Examples of isometries include translations, rotations, reflections, and glide reflections. The set of all isometries of a plane under composition of functions forms a group.

Example 1.2.5: Isometries of a plane are not an Abelian group

To show that the group T of isometries of a plane is not Abelian, we only need to find two isometries ϕ and θ such that $\phi \circ \theta \neq \theta \circ \phi$. The functions $\phi(x, y) = (-x, y)$ (reflection across the y -axis) and $\theta(x, y) = (-y, x)$ (rotation by $\pi/2$ about the origin) are such isometries since $\phi \circ \theta(x, y) = \phi(-y, x) = (y, x)$, but $\theta \circ \phi(x, y) = \theta(-x, y) = (y, -x)$, which implies that $\phi \circ \theta \neq \theta \circ \phi$. Hence, T is not an Abelian group under composition of functions.

Example 1.2.6: Show that the subset S of $M_n(\mathbb{R})$ consisting of all invertible $n \times n$ matrices under matrix multiplication is a group but not Abelian.

We start by showing that S is closed under matrix multiplication. Let A and B be in S , so that both A^{-1} and B^{-1} exist and $AA^{-1} = BB^{-1} = I_n$. Then

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$

so that AB is invertible and consequently $AB \in S$.

Since matrix multiplication is associative and I_n acts as the identity element, and since each element of S has an inverse by definition of S , we see that S is indeed a group. $\square$

To prove that S is not Abelian, we only need to find two invertible matrices A and B

such that $AB \neq BA$. For example, let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. Then we have:

$$AB = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

so that $AB \neq BA$. Hence, S is not an Abelian group under matrix multiplication. $\square$

1.2.1 Elementary Properties of Groups

Theorem 1.2.1 (Left and Right Cancellation Laws): If G is a group with binary operation $*$, then the **left and right cancellation laws** hold in G , that is,

$$a * b = a * c \implies b = c \quad \text{and} \quad b * a = c * a \implies b = c$$

Proof:

Let $a * b = a * c$. Then by $\mathcal{G}_3$, there exists a' , and

$$\begin{aligned} a' * (a * b) &= a' * (a * c) \\ (a' * a) * b &= (a' * a) * c && \text{(associativity)} \\ e * b &= e * c && \text{(identity element)} \\ b &= c && \text{(identity element)} \end{aligned}$$

Similarly, from $b * a = c * a$, we can deduce that $b = c$ upon multiplication on the right by a' and use of the axioms for a group. $\square$

Theorem 1.2.2 (Uniqueness of Solutions to Linear Equations in Groups): If $\langle G, * \rangle$ is a group, and if a and b are elements of G , then the linear equations $a * x = b$ and $y * a = b$ have unique solutions x and y in G

Proof:

Let a and b be elements of G . By $\mathcal{G}_3$, there is an element a' such that $a * a' = e$. Then we have:

$$\begin{aligned} a * x &= b \\ (a * a') * x &= a' * b && \text{(multiplying on the left by } a') \\ e * x &= a' * b && \text{(identity element)} \\ x &= a' * b \end{aligned}$$

so that $x = a' * b$ is a solution to the equation $a * x = b$. To show that this solution is unique, suppose that x' is another solution to the equation, so that $a * x' = b$. Then we have:

$$\begin{aligned} a * x' &= b \\ (a * a') * x' &= a' * b && \text{(multiplying on the left by } a') \\ e * x' &= a' * b && \text{(identity element)} \\ x' &= a' * b \end{aligned}$$

so that $x' = x$, and the solution to the equation $a * x = b$ is unique. A similar argument shows that the solution to the equation $y * a = b$ is also unique. $\square$

Theorem 1.2.3 (Uniqueness of Identity and Inverse Elements in Groups): If $\langle G, * \rangle$ is a group, there is only one element e in G such that

$$\forall x \in G, \quad e * x = x * e = x$$

Likewise, for each $a \in G$, there is only one element a' in G such that

$$a * a' = a' * a = e$$

Proof:

Let e and e' be elements of G such that for all $x \in G$, we have $e * x = x * e = x$ and $e' * x = x * e' = x$. Then we have:

$$\begin{aligned} e &= e * e' && \text{(since } e' \text{ is an identity element)} \\ &= e' && \text{(since } e \text{ is an identity element)} \end{aligned}$$

so that there is only one identity element in G .

Let a be an element of G , and let a' and a'' be elements of G such that $a * a' = a' * a = e$ and $a * a'' = a'' * a = e$. Then we have:

$$\begin{aligned} a' &= a' * e && \text{(identity element)} \\ &= a' * (a * a'') && \text{(since } a * a'' = e) \\ &= (a' * a) * a'' && \text{(associativity)} \\ &= e * a'' && \text{(since } a' * a = e) \\ &= a'' && \text{(identity element)} \end{aligned}$$

so that there is only one inverse for each element of G . $\square$

Corollary 1.2.4: Let G be a group. For all $a, b \in G$, we have $(a * b)' = b' * a'$

Proof:

Let a and b be elements of G . Then we have:

$$\begin{aligned} (a * b) * (b' * a') &= a * (b * b') * a' && \text{(associativity)} \\ &= a * e * a' && \text{(since } b * b' = e) \\ &= a * a' && \text{(identity element)} \\ &= e \end{aligned}$$

and

$$\begin{aligned} (b' * a') * (a * b) &= b' * (a' * a) * b && \text{(associativity)} \\ &= b' * e * b && \text{since } a' * a = e \\ &= b' * b && \text{(identity element)} \\ &= e \end{aligned}$$

so that $b' * a'$ is the inverse of $a * b$. Since the inverse of an element in a group is unique, we conclude that $(a * b)' = b' * a'$. $\square$

Note:-

We remark that binary algebraic structures with weaker axioms than those for a group have also been studied quite extensively. Of these weaker structures, the **semigroup**, a set with an associative binary operation, has had the most attention. A **monoid** is a semigroup that has an identity element for the binary operation. Note that every group is both a semigroup and a monoid.

Weakening the group axioms

We can weaken the group axioms by only requiring that there is a left identity element e such that for all $x \in G$, we have $e * x = x$, and that for each $a \in G$, there is a left inverse a' such that $a' * a = e$. However, it turns out that these weaker axioms are actually equivalent to the original group axioms. That is, if a set with a binary operation satisfies the weaker axioms, then it also satisfies the original group axioms, and hence is a group. To see this, let G be a set with a binary operation $*$ such that there is a left identity element e and that for each $a \in G$, there is a left inverse a' such that $a' * a = e$. Then we have:

$$\begin{aligned} a * a' &= a * (a' * a) * a' && \text{(associativity)} \\ &= (a * a') * a * a' && \text{(associativity)} \\ &= e * a * a' && \text{(since } a' * a = e) \\ &= a * a' && \text{(identity element)} \end{aligned}$$

so that $a * a' = e$. Similarly, we can show that $a' * a = e$ for all $a \in G$. Hence, the weaker axioms imply the original group axioms, and we conclude that the weaker axioms are equivalent to the original group axioms.

1.2.2 Group Isomorphisms

Definition 1.2.5: Group Isomorphism

Let $\langle G_1, *_1 \rangle$ and $\langle G_2, *_2 \rangle$ be groups and $f : G_1 \rightarrow G_2$. We say that f is a **group isomorphism** if the following conditions are satisfied:

1. The function f is a bijection (one-to-one and onto) from G_1 to G_2 .
2. (**Homomorphism property**): For all $a, b \in G_1$, $f(a *_1 b) = f(a) *_2 f(b)$.

Notation for isomorphisms

If there is a group isomorphism from G_1 to G_2 , we say that G_1 and G_2 are **isomorphic** groups, and we write $G_1 \cong G_2$.

Example 1.2.7: The set of even integers, $2\mathbb{Z}$, forms a group under addition. Prove that $\mathbb{Z}$ and $2\mathbb{Z}$ are isomorphic groups.

Let $f : \mathbb{Z} \rightarrow 2\mathbb{Z}$ be a bijection defined by $f(n) = 2n$.

To verify the first condition, let $a, b \in \mathbb{Z}$ and $f(a) = f(b)$. Then $2a = 2b$, which implies that $a = b$, so f is one-to-one. To show that f is onto, let $y \in 2\mathbb{Z}$. Since y is even, $\exists c \in \mathbb{Z}$ s.t. $y = 2c$. Therefore, $y = 2c = f(c)$, so f maps onto $2\mathbb{Z}$.

To verify the second condition, i.e., the homomorphism property, let $a, b \in \mathbb{Z}$. Then

$$f(a + b) = 2(a + b) = 2a + 2b = f(a) + f(b)$$

which shows that f is a group homomorphism. Since f is a bijection and a homomor-

phism, we conclude that f is a group isomorphism, and hence $\mathbb{Z}$ and $2\mathbb{Z}$ are isomorphic groups. $\square$

1.3 Abelian Examples

Definition 1.3.1: Addition Modulo n

Let $\mathbb{Z}_n = \{0, 1, 2, 3, \dots, n-1\}$. We define the operation $+_n$ on $\mathbb{Z}_n$ by

$$a +_n b = \begin{cases} a + b & \text{if } a + b < n \\ a + b - n & \text{if } a + b \geq n \end{cases}$$

This operation is called **addition modulo n** .

For any $a, b \in \mathbb{Z}_n$, we have

$$\begin{aligned} 0 &\leq a + b \leq 2n - 2 \\ \text{so, } 0 &\leq a +_n b \leq n - 1 \end{aligned}$$

Thus, $+_n$ is indeed an operation on $\mathbb{Z}_n$.

Example 1.3.1: Prove that $\langle \mathbb{Z}_n, +_n \rangle$ is an Abelian group

The identity element for $+_n$ is 0, since for any $a \in \mathbb{Z}_n$, we have $0 +_n a = a$ and $a +_n 0 = a$. The inverse of any element a in $\mathbb{Z}_n$ is $n - a$, since we have $a +_n (n - a) = 0$ and $(n - a) +_n a = 0$.

Since addition of integers is commutative, we have that $+_n$ is commutative.

To prove the associativity is the tricky part of this problem. Let $a, b, c \in \mathbb{Z}_n$. Then we have:

$$\begin{aligned} (a +_n b) +_n c &= \begin{cases} (a + b) +_n c & \text{if } a + b < n \\ (a + b - n) +_n c & \text{if } a + b \geq n \end{cases} \\ &= \begin{cases} a + (b + c) & \text{if } a + b < n \text{ and } b + c < n \\ a + (b + c - n) & \text{if } a + b < n \text{ and } b + c \geq n \\ (a + b - n) + c & \text{if } a + b \geq n \text{ and } (a + b - n) + c < n \\ (a + b - n) + (c - n) & \text{if } a + b \geq n \text{ and } (a + b - n) + c \geq n \end{cases} \\ &= \begin{cases} a + (b + c) & \text{if } a + b < n \text{ and } b + c < n \\ a + (b + c - n) & \text{if } a + b < n \text{ and } b + c \geq n \\ a + (b + c) - n & \text{if } a + b \geq n \text{ and } (a + b - n) + c < n \\ a + (b + c - n) - n & \text{if } a + b \geq n \text{ and } (a + b - n) + c \geq n \end{cases} \\ &= a +_n (b +_n c) \end{aligned}$$

so that $+_n$ is associative. Hence, $\langle \mathbb{Z}_n, +_n \rangle$ is an Abelian group. $\square$

Alternative proof of associativity: Let $a, b, c \in \mathbb{Z}_n$. Then we have:

$$\begin{aligned}
 (a +_n b) +_n c &= \left[\{(a + b) \pmod n\} + c \right] \pmod n \\
 &= (a + b + c) \pmod n \\
 &= (a + (b + c)) \pmod n \\
 &= \left[a + \{(b + c) \pmod n\} \right] \pmod n \\
 &= a +_n (b +_n c)
 \end{aligned}$$

Thus, $+_n$ is associative, and $\langle \mathbb{Z}_n, +_n \rangle$ is an Abelian group. $\square$

Note:-

In the alternative proof, we're using this fact:

$$(x \pmod n + c) \pmod n = (x + c) \pmod n$$

Proof:

Let $x \in \mathbb{Z}$. Then by the division algorithm, $\exists q$ s.t. $x = qn + r$, where $0 \leq r < n$. Then we have:

$$\begin{aligned}
 (x \pmod n + c) \pmod n &= (r + c) \pmod n \\
 &= ((qn + r) + c) \pmod n \\
 &= (x + c) \pmod n
 \end{aligned}$$

Lemma 1.3.1: *Completely different operations on sets can still define isomorphic groups.*

Example 1.3.2: Prove that $\mathbb{Z}_2 = \{0, 1\}$ is isomorphic with $\{1, -1\}$ under multiplication

Let $f : \mathbb{Z}_2 \rightarrow \{1, -1\}$ be defined by $f(0) = 1$ and $f(1) = -1$. Then f is a bijection since $f(0) \neq f(1)$ and for each element in $\{1, -1\}$, there is an element in $\mathbb{Z}_2$ that maps to it. To verify the homomorphism property, let $a, b \in \mathbb{Z}_2$. Then we have:

$$\begin{aligned}
 f(a +_2 b) &= f((a + b) \pmod 2) \\
 &= (-1)^{(a+b) \pmod 2} \\
 &= (-1)^a (-1)^b \\
 &= f(a)f(b)
 \end{aligned}$$

so that f is a group homomorphism. Since f is a bijection and a homomorphism, we conclude that f is a group isomorphism, and hence $\mathbb{Z}_2$ and $\{1, -1\}$ are isomorphic groups. $\square$

Example 1.3.3: Klein 4-group

The group $\mathbb{Z}_2 \times \mathbb{Z}_2$ is called the **Klein 4-group** and is denoted by V . The elements of V are $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$, and the operation on V is defined by

$$(a, b) + (c, d) = ((a + c) \pmod 2, (b + d) \pmod 2)$$

for all $(a, b), (c, d) \in V$. The identity element for this operation is $(0, 0)$, and the inverse

of any element (a, b) is itself since $(a, b) + (a, b) = (0, 0)$. Since addition of integers is commutative, we have that the operation on V is commutative. It can be verified that the operation on V is associative as well. Hence, $\langle V, + \rangle$ is an Abelian group.

Note:-

Note that Klein 4-group V is not isomorphic to $\mathbb{Z}_4$ since $\mathbb{Z}_4$ has an element of order 4 (that is, you need to add it to itself 4 times to get back to the identity), but in $V = \mathbb{Z}_2 \times \mathbb{Z}_2$, every non-identity element has order 2.

However, V is isomorphic to the group of symmetries of a non-square rectangle, which consists of the identity symmetry, the reflection across the vertical axis, the reflection across the horizontal axis, and the rotation by π about the center of the rectangle.

Example 1.3.4: $\langle \mathbb{R}_{2\pi}, +_{2\pi} \rangle$

In the group $\langle \mathbb{R}_{2\pi}, +_{2\pi} \rangle$, we are essentially equating 0 with 2π in the sense that if a and b add to give 2π , we know that $a +_{2\pi} b = 0$. Geometrically, we can think of this as taking a circle of radius 1.

1.3.1 Complex Numbers and the Circle Group

Definition 1.3.2: Complex Numbers

The set $\mathbb{C}$ of **complex numbers** is defined by

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$$

where i is the imaginary unit satisfying $i^2 = -1$. A complex number can be regarded as a point in the Euclidean plane, where the point with Cartesian coordinates (a, b) is labeled $a + bi$.

Multiplication of Complex Numbers

The product of $z_1 = a + bi$ and $z_2 = c + di$ is defined as

$$z_1 z_2 = (a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

It is routine to check that multiplication of complex numbers satisfies commutativity, associativity, and distributivity.

Absolute Value and Polar Form

The **absolute value** of a complex number $a + bi$ is defined by

$$|a + bi| = \sqrt{a^2 + b^2}$$

This is the distance from $a + bi$ to the origin. We can describe a complex number z in **polar form** as

$$z = |z|(\cos \theta + i \sin \theta)$$

where θ is the angle measured counterclockwise from the positive x -axis to the vector from 0 to z .

Euler's Formula

Euler's formula states that $e^{i\theta} = \cos \theta + i \sin \theta$. Using this formula, we can express z in polar form as $z = |z|e^{i\theta}$.

Geometric Interpretation of Complex Multiplication

If $z_1 = |z_1|e^{i\theta_1}$ and $z_2 = |z_2|e^{i\theta_2}$, then

$$z_1 z_2 = |z_1||z_2|e^{i(\theta_1+\theta_2)} = |z_1||z_2|[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Geometrically, we multiply complex numbers by multiplying their absolute values and adding their polar angles.

Example 1.3.5: Find all solutions in $\mathbb{C}$ of the equation $z^2 = i$

Writing the equation $z^2 = i$ in polar form, we obtain

$$|z|^2(\cos 2\theta + i \sin 2\theta) = 1(0 + i)$$

Thus $|z|^2 = 1$, so $|z| = 1$. The angle θ for z must satisfy $\cos 2\theta = 0$ and $\sin 2\theta = 1$. Consequently, $2\theta = \frac{\pi}{2} + n(2\pi)$, so $\theta = \frac{\pi}{4} + n\pi$ for an integer n . The values of n yielding values θ where $0 \leq \theta < 2\pi$ are 0 and 1, yielding $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$. Our solutions are

$$z_1 = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}(1 + i) \quad \text{and} \quad z_2 = \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} = \frac{-1}{\sqrt{2}}(1 + i)$$

□

Example 1.3.6: Find all solutions in $\mathbb{C}$ of the equation $z^4 = -16$

Writing the equation in polar form, we get

$$|z|^4(\cos 4\theta + i \sin 4\theta) = 16(-1 + 0i)$$

Here, $|z|^4 = 16$, so $|z| = 2$. Also, $\cos 4\theta = -1$ and $\sin 4\theta = 0$, and so $4\theta = \pi + n(2\pi)$, so $\theta = (\pi/4) + n(\pi/2)$ for $n \in \mathbb{Z}$. Thus, for $0 \leq \theta < 2\pi$, we get the values $\theta = \pi/4, 3\pi/4, 5\pi/4$, and $7\pi/4$.

Thus, the solutions of $z^4 = -16$ are

$$\sqrt{2}(1 + i), \quad \sqrt{2}(-1 + i), \quad \sqrt{2}(-1 - i), \quad \sqrt{2}(1 - i)$$

Definition 1.3.3: The Circle Group

Let $U = \{z \in \mathbb{C} \mid |z| = 1\}$, so that U is the circle in the Euclidean plane with center at the origin and radius 1. We call $\langle U, \cdot \rangle$ the **circle group**.

Theorem 1.3.2 (The Circle Group is Abelian): $\langle U, \cdot \rangle$ is an Abelian group.

Proof:

We first check that U is closed under multiplication. Let $z_1, z_2 \in U$. Then $|z_1| = |z_2| = 1$, which implies that $|z_1 z_2| = |z_1||z_2| = 1$, showing $z_1 z_2 \in U$. □

Since multiplication of complex numbers is associative and commutative in general, multiplication in U is also associative and commutative, which verifies $\mathcal{G}_1$ and the condition for

Abelian. □

The number $1 \in U$ is the identity, verifying condition $\mathcal{G}_2$. □

For each $a + bi \in U$, we have

$$(a + bi)(a - bi) = a^2 - (bi)^2 = a^2 + b^2 = |a + bi|^2 = 1$$

So the inverse of $a + bi$ is $a - bi$, which verifies condition $\mathcal{G}_3$. Thus, U is an Abelian group under multiplication. □

Isomorphism between U and $\mathbb{R}_{2\pi}$

We can relabel points in U as points in $\mathbb{R}_{2\pi}$ by simply relabeling z as θ where $0 \leq \theta < 2\pi$. Let $f : U \rightarrow \mathbb{R}_{2\pi}$ be given by $f(z) = \theta$ according to this relabeling. Then for $z_1, z_2 \in U$, we have $f(z_1 z_2) = f(z_1) +_{2\pi} f(z_2)$ since multiplying in U simply adds the corresponding angles: if $z_1 \leftrightarrow \theta_1$ and $z_2 \leftrightarrow \theta_2$, then $z_1 \cdot z_2 \leftrightarrow (\theta_1 +_{2\pi} \theta_2)$.

1.3.2 Roots of Unity

Definition 1.3.4: Roots of Unity

The elements of the set $U_n = \{z \in \mathbb{C} \mid z^n = 1\}$ are called the n^{th} **roots of unity**. The elements of this set are the numbers

$$e^{\left(m \frac{2\pi}{n}\right)i} = \cos\left(m \frac{2\pi}{n}\right) + i \sin\left(m \frac{2\pi}{n}\right) \quad \text{for } m = 0, 1, 2, \dots, n-1$$

The elements have absolute value 1 and are evenly spaced around the unit circle in the complex plane.

Lemma 1.3.3: *The n^{th} roots of unity form an Abelian group under multiplication.*

Proof:

Let z_1 and z_2 be n^{th} roots of unity. Then $z_1^n = z_2^n = 1$. We have

$$(z_1 z_2)^n = z_1^n z_2^n = 1 \cdot 1 = 1$$

so that $z_1 z_2$ is also an n^{th} root of unity, and hence the set of n^{th} roots of unity is closed under multiplication.

Since multiplication of complex numbers is associative, we have that multiplication in the set of n^{th} roots of unity is associative, which verifies $\mathcal{G}_1$.

The number 1 is an n^{th} root of unity and serves as the identity element, verifying $\mathcal{G}_2$.

For each n^{th} root of unity z , we have $z^{-1} = z^*$, where z^* is the complex conjugate of z , since we have

$$z \cdot z^* = |z|^2 = 1$$

so that z^* is the inverse of z , which verifies $\mathcal{G}_3$. Thus, the n^{th} roots of unity form a group under multiplication.

Since multiplication of complex numbers is commutative, we have that multiplication in the set of n^{th} roots of unity is commutative, which verifies the condition for Abelian. Thus, the n^{th} roots of unity form an Abelian group under multiplication. □

Isomorphism between U_n and $\mathbb{Z}_n$

We can relabel the n^{th} roots of unity as elements of $\mathbb{Z}_n$ by relabeling $e^{\left(m \frac{2\pi}{n}\right)i}$ as m . Let $f :$

$U_n \rightarrow \mathbb{Z}_n$ be given by $f\left(e^{\left(m\frac{2\pi}{n}\right)i}\right) = m$. Then for $z_1, z_2 \in U_n$, we have $f(z_1 z_2) = f(z_1) +_n f(z_2)$ since multiplying in U_n simply adds the corresponding angles: if $z_1 \leftrightarrow m_1$ and $z_2 \leftrightarrow m_2$, then $z_1 \cdot z_2 \leftrightarrow (m_1 +_n m_2)$.

1.4 Nonabelian Examples

1.4.1 Notation and Terminology

Algebraists usually do not use the symbol $*$ to denote the group operation. Instead, they often use juxtaposition, so that $a * b$ is denoted by ab . In this case, the identity element is denoted by 1 or e , and the inverse of an element a is denoted by a^{-1} instead of a' . When we write ab , we mean the product of a and b under the group operation. The notation $a + b$ is also often used to denote the group operation, especially when the group is Abelian. When the group operation is denoted by $+$, the identity element is denoted by 0, and the inverse of an element a is denoted by $-a$. When we write $a + b$, we mean the sum of a and b under the group operation.

Summary

$\cdot$ Notation	$+$ Notation
May or may not be Abelian	Abelian
e	0
a^{-1}	$-a$
ab	$a + b$
a^k	ka
a^{-k}	$-ka$
$a^0 = 1$	$0a = 0$
$a^{n+m} = a^n a^m$	$(n + m)a = na + ma$
$(a^n)^m = a^{nm}$	$m(na) = (mn)a$

Definition 1.4.1: Order of group

The **order** of a group G , denoted by $|G|$, is the number of elements in G .

1.4.2 Permutation Groups

Definition 1.4.2: Permutation

A **permutation** of a set A is a bijection $\sigma : A \rightarrow A$.

Definition 1.4.3: Permutation multiplication

Function composition $\circ$ is a binary operation on the collection of all permutations of a set A . We call this operation **permutation multiplication**.

Let σ and τ be permutations of A . The composite function $\sigma \circ \tau$ defined schematically by

$$A \xrightarrow{\tau} A \xrightarrow{\sigma} A$$

gives a bijection from A to A , and hence is a permutation of A .

Lemma 1.4.1: *The permutation multiplication is bijective.*

Proof:

Let σ and τ be permutations of A . Let's first show that $\sigma\tau$ is injective. If $(\sigma\tau)(a_1) = (\sigma\tau)(a_2)$, then $\sigma(\tau(a_1)) = \sigma(\tau(a_2))$. Since σ is injective, we have $\tau(a_1) = \tau(a_2)$. And since τ is injective, we conclude that $a_1 = a_2$, so $\sigma\tau$ is injective.

To show that $\sigma\tau$ is surjective, let $b \in A$. Since σ is surjective, there exists $a \in A$ such that $\sigma(a) = b$. Since τ is surjective, there exists $c \in A$ such that $\tau(c) = a$. Thus, we have $(\sigma\tau)(c) = \sigma(\tau(c)) = \sigma(a) = b$, so $\sigma\tau$ is surjective. Hence, $\sigma\tau$ is bijective. $\square$

Example 1.4.1

Suppose that

$$A = \{1, 2, 3, 4, 5\}$$

and that σ is the permutation given by the following diagram:

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 4 & 2 & 5 & 3 & 1 \end{array}$$

and τ is the permutation given by the following diagram:

$$\begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 3 & 5 & 4 & 2 & 1 \end{array}$$

Then we can write σ and τ in two-row notation as

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}$$

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}$$

Then the convolution can be written as:

$$\begin{aligned} \sigma\tau &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ \sigma(3) & \sigma(5) & \sigma(4) & \sigma(2) & \sigma(1) \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 3 & 2 & 4 \end{pmatrix} \end{aligned}$$

Theorem 1.4.2: *Let A be a nonempty set, and let S_A be the collection of all permutations of A . Then S_A is a group under permutation multiplication.*

Proof:

We first check that S_A is closed under permutation multiplication. Let σ and τ be permutations of A . Then $\sigma\tau$ is a permutation of A , so $\sigma\tau \in S_A$.

Since function composition is associative, we have that permutation multiplication is associative, which verifies $\mathcal{G}_1$.

The permutation ι such that $\iota(a) = a$ for all $a \in A$ acts as identity. Therefore, $\mathcal{G}_2$ is satisfied. For a permutation σ , the inverse function σ^{-1} is the permutation that reverses the direction of the mapping σ .

The existence of exactly one such element σ^{-1} is guaranteed by the fact that σ is a bijection. Thus, $\mathcal{G}_3$ is satisfied, and S_A is a group under permutation multiplication. $\square$

Definition 1.4.4: Symmetric Group

Let A be the finite set $\{1, 2, \dots, n\}$. The group of all permutations of A is the **symmetric group on n letters**, and is denoted by S_n .

Note:-

S_n has $n!$ elements, where $n! = n(n-1)(n-2) \cdots (3)(2)(1)$.

Example 1.4.2

Let $\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$. Then $\sigma\tau(1) = \sigma(1) = 2$, but $\tau\sigma(1) = \tau(2) = 3$.

Thus $\sigma\tau \neq \tau\sigma$, so S_3 is not Abelian. We have seen that any group with at most four elements is Abelian. Furthermore, up to isomorphism, the Abelian group $\mathbb{Z}_5$ is the only group of order 5. Thus S_3 is the smallest group which is not Abelian.

1.4.3 Disjoint Cycles

There is a more efficient way of specifying the action of a permutation. In the two-row notation, we list each number 1 through n twice. Disjoint cycle notation allows us to write the permutation using each number only once.

Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 6 & 2 & 5 & 1 \end{pmatrix}$. To write in disjoint cycle notation, we start by writing

$$(1$$

We see that $\sigma(1) = 3$, so we write:

$$(1, 3$$

Now σ maps 3 to 6, so we write:

$$(1, 3, 6$$

Our permutation maps 6 back to 1, so we place a parenthesis after 6:

$$(1, 3, 6)$$

This is called a **cycle** because when we apply σ repeatedly, we cycle through the numbers 1, 3, and 6. A cycle containing exactly k numbers is called a k -cycle. So $(1, 3, 6)$ is a 3-cycle.

We start another cycle for the remaining elements: σ maps 2 to 4 and 4 maps back to 2, giving the 2-cycle

$$(2, 4)$$

Since 5 maps to itself, we omit it. Thus in disjoint cycle notation,

$$\sigma = (1, 3, 6)(2, 4)$$

A 2-cycle is called a **transposition**. A collection of cycles is said to be **disjoint** if no entry is in more than one cycle. Disjoint cycles commute with each other.

Example 1.4.3

In disjoint cycle notation, $\sigma \in S_9$ is written as $(1, 5, 2, 7)(3, 4, 9)$. Reading off the disjoint cycle notation, $\sigma(1) = 5, \sigma(5) = 2, \sigma(2) = 7, \sigma(7) = 1, \sigma(3) = 4, \sigma(4) = 9, \sigma(9) = 3$. Since 6 and 8 do not appear in either cycle, $\sigma(6) = 6$ and $\sigma(8) = 8$. Thus in two-row notation,

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 5 & 7 & 4 & 9 & 1 & 6 & 2 & 8 & 3 \end{pmatrix}$$

Lemma 1.4.3: *Disjoint cycles commute with each other.*

Proof:

Let σ and τ be disjoint cycles. Let $a \in \{1, 2, \dots, n\}$. If a is not in either cycle, then $\sigma\tau(a) = a = \tau\sigma(a)$. If a is in σ but not in τ , then $\tau(a) = a$, so $\sigma\tau(a) = \sigma(a)$ and $\tau\sigma(a) = \tau(\sigma(a)) = \sigma(a)$, so $\sigma\tau(a) = \tau\sigma(a)$. If a is in τ but not in σ , then $\sigma(a) = a$, so $\sigma\tau(a) = \sigma(\tau(a)) = \tau(a)$ and $\tau\sigma(a) = \tau(a)$, so $\sigma\tau(a) = \tau\sigma(a)$. Thus, for all a , we have $\sigma\tau(a) = \tau\sigma(a)$, so $\sigma\tau = \tau\sigma$. $\square$

Example 1.4.4: Let $\sigma = (1, 5, 3, 2, 6)$ and $\tau = (1, 2, 4, 3, 6)$ in S_6 . Find $\sigma\tau$ in disjoint cycle notation without writing σ and τ in two-row notation.

$$\sigma\tau = (1, 5, 3, 2, 6)(1, 2, 4, 3, 6)$$

Since the operation is function composition, we see that the cycle τ on the right sends 1 to 2 and then the cycle on the left sends 2 to 6. So, $\sigma\tau(1) = 6$ and so we can start writing

$$\sigma\tau = (1, 6$$

Now we see that τ maps 6 to 1 and σ maps 1 to 5. So,

$$\sigma\tau = (1, 6, 5$$

Next, $\tau(5) = 5$ and $\sigma(5) = 3$, so

$$\sigma\tau = (1, 6, 5, 3$$

Next, $\tau(3) = 6$ and $\sigma(6) = 1$, so

$$\sigma\tau = (1, 6, 5, 3, 1$$

Since we have come back to 1, we close the parenthesis:

$$\sigma\tau = (1, 6, 5, 3)$$

Next, we look for the smallest number not in the cycle we just wrote. The smallest number not in the cycle is 2. We have $\tau(2) = 4$ and $\sigma(4) = 4$, so we write

$$\sigma\tau = (1, 6, 5, 3)(2, 4$$

Next, $\tau(4) = 3$ and $\sigma(3) = 2$. Since we have come back to 2, we close the parenthesis:

$$\sigma\tau = (1, 6, 5, 3)(2, 4)$$

Finally, we check if there are any numbers not in either cycle. The only number not in either cycle is 6, but 6 maps to itself, so we can omit it.

Thus, $\sigma\tau = (1, 6, 5, 3)(2, 4)$.

Example 1.4.5: Compute the inverse of $\sigma = (1, 5, 7)(3, 8, 2, 4, 6)$

In general, for a group $(ab)^{-1} = b^{-1}a^{-1}$, so

$$\sigma^{-1} = (3, 8, 2, 4, 6)^{-1}(1, 5, 7)^{-1}$$

The inverse of a cycle is simply written backward:

$$\sigma^{-1} = (6, 4, 2, 8, 3)(7, 5, 1)$$

However, since disjoint cycles commute, we can start each cycle with any entry in the cycle. So we could write

$$\sigma^{-1} = (7, 5, 1)(6, 4, 2, 8, 3)$$

or

$$\sigma^{-1} = (1, 7, 5)(2, 8, 3, 6, 4)$$

Note:-

The inverse of a cycle is simply the cycle written backward. For example, $(1, 5, 7)^{-1} = (7, 5, 1)$.

Example 1.4.6:

(a) Verify that the six matrices form a group under matrix multiplication.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) What group discussed in this section is this group isomorphic to?

- (a) We first check that the set of matrices is closed under matrix multiplication. Let A and B be any two of the six matrices. Then AB is a matrix with exactly one entry of 1 in each row and column, and all other entries are 0. Thus, AB is one of the six matrices, so the set of matrices is closed under matrix multiplication.

Since matrix multiplication is associative, we have that matrix multiplication in this set is associative, which verifies $\mathcal{G}_1$.

The identity matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ serves as the identity element, verifying $\mathcal{G}_2$.

For each of the six matrices, we can find an inverse matrix in the set such that their product is the identity matrix. Thus, $\mathcal{G}_3$ is satisfied, and the set of matrices forms a group under matrix multiplication.

- (b) This group is isomorphic to S_3 , the symmetric group on three letters, since the six matrices correspond to the six permutations of three elements. For example, the matrix

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

corresponds to the permutation that sends 1 to 2, 2 to 3, and 3 to 1, which is the cycle $(1, 2, 3)$ in S_3 . The other matrices correspond to the other permutations in S_3 in a similar way. Thus, the group of matrices is isomorphic to S_3 .

1.4.4 The Dihedral Group

To define the dihedral group, we first set up coordinates. Let P_n denote the regular n -gon inscribed in the unit circle centered at the origin. We label the vertices with elements of $\mathbb{Z}_n$: vertex k is placed at angle $\frac{2\pi k}{n}$ measured counterclockwise from the positive x -axis. So vertex 0 sits at $(1, 0)$, vertex 1 at $\left(\cos \frac{2\pi}{n}, \sin \frac{2\pi}{n}\right)$, and so on. The edges of P_n are the line segments between vertices k and $k +_n 1$ for $0 \leq k \leq n - 1$.

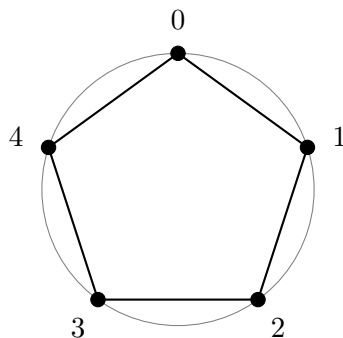


Figure 1.4.1: The regular pentagon P_5 with vertex labels in $\mathbb{Z}_5$

Definition 1.4.5: Dihedral Group

The **dihedral group** D_n is the set of all bijections $\phi : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ such that for any $i, j \in \mathbb{Z}_n$, the line between vertices i and j is an edge of P_n if and only if the line between $\phi(i)$ and $\phi(j)$ is an edge of P_n .

Note:-

In other words, the elements of D_n are the symmetries of the regular n -gon: the bijections of the vertex set that preserve the edge structure.

Theorem 1.4.4: For any $n \geq 3$, $\langle D_n, \circ \rangle$ is a group.

Proof:

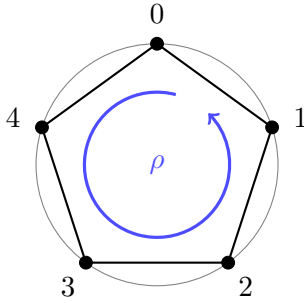
Let $\phi, \theta \in D_n$ and suppose the line between vertices i and j is an edge of P_n . Since $\theta \in D_n$, the line between $\theta(i)$ and $\theta(j)$ is an edge. Since $\phi \in D_n$, the line between $\phi(\theta(i))$ and $\phi(\theta(j))$ is an edge. The converse direction follows similarly. Since the composition of bijections is a bijection, $\phi \circ \theta \in D_n$.

Function composition is associative ($\mathcal{G}_1$). The identity function $\iota(k) = k$ preserves all edges ($\mathcal{G}_2$). For any $\phi \in D_n$, the inverse function ϕ^{-1} is also a bijection preserving edges ($\mathcal{G}_3$), since ϕ maps edges to edges if and only if ϕ^{-1} does. $\square$

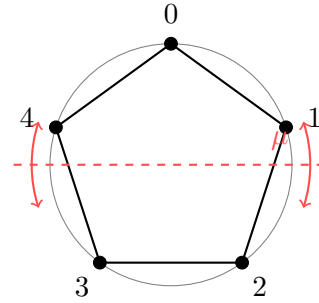
The generators ρ and μ

Two fundamental elements of D_n are **rotation** ρ and **reflection** μ :

- $\rho : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ defined by $\rho(k) = k +_n 1$ (rotation by $2\pi/n$ counterclockwise). This sends each vertex to the next one around the circle.
- $\mu : \mathbb{Z}_n \rightarrow \mathbb{Z}_n$ defined by $\mu(k) = -k = n - k$ for $k > 0$ and $\mu(0) = 0$ (reflection across the x -axis, i.e., the line through vertex 0 and the center).



Rotation $\rho(k) = k +_n 1$



Reflection $\mu(k) = -k$

Figure 1.4.2: The generators ρ (rotation) and μ (reflection) of the dihedral group D_5

Example 1.4.7: In D_5

ρ sends $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$. In cycle notation, $\rho = (0, 1, 2, 3, 4)$.

μ fixes vertex 0 and swaps $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$. In cycle notation, $\mu = (1, 4)(2, 3)$.

Theorem 1.4.5: For $n \geq 3$, the dihedral group D_n has order $2n$, and

$$D_n = \{\iota, \rho, \rho^2, \dots, \rho^{n-1}, \mu, \mu\rho, \mu\rho^2, \dots, \mu\rho^{n-1}\}$$

Proof:

We first show that D_n can have at most $2n$ elements. Any $\phi \in D_n$ maps vertex 0 to some vertex y (n choices). Since y is connected to exactly two vertices by edges, vertex 1 (which is adjacent to 0) must map to one of these two neighbors (2 choices). Once the images of vertices 0 and 1 are fixed, the rest are determined (each subsequent vertex is adjacent to the previous, so the image of vertex k is forced). Thus $|D_n| \leq n \times 2 = 2n$.

Now we need to show that $|D_n| = 2n$. For that, we need to show that no two of the functions $\rho^0, \rho^1, \rho^2, \dots, \rho^{n-1}, \mu, \mu\rho, \mu\rho^2, \dots, \mu\rho^{n-1}$ are the same. Let's assume $\rho^k = \rho^r$ for some $0 \leq k, r \leq n-1$ ($k, r \in \mathbb{Z}$). Then

$$\begin{aligned} \rho^k(m) &= \rho^r(m), & [0 \leq m \leq n-1] \\ k +_n m &= r +_n m \\ k &= r \end{aligned}$$

So, no two of the ρ^k 's are the same.

Next, we show that no ρ^k is equal to any $\mu\rho^r$. If $\rho^k = \mu\rho^r$, then $\mu = \rho^{k-r}$, which is a rotation. But μ is a reflection, so this is impossible.

Finally, we show that no two of the $\mu\rho^k$'s are the same. If $\mu\rho^k = \mu\rho^r$, then $\rho^k = \rho^r$, which we have already shown implies $k = r$. Thus, all $2n$ functions are distinct, so $|D_n| = 2n$. $\square$

Definition 1.4.6: Standard Form

Every element of D_n can be written uniquely as ρ^k or $\mu\rho^k$ for some $0 \leq k \leq n-1$. This is the **standard form** of an element of D_n .

To do arithmetic in D_n , the crucial relation is:

Theorem 1.4.6: In D_n for $n \geq 3$, we have $\rho\mu = \mu\rho^{n-1}$, or equivalently, $\rho^k\mu = \mu\rho^{n-k}$ for all k .

Proof:

We derive $\mu\rho\mu = \rho^{n-1}$. Each application of μ reverses the orientation (CW $\leftrightarrow$ CCW), so $\mu\rho\mu$ reverses twice, giving a rotation. We determine which rotation by evaluating at 0:

$$\mu\rho\mu(0) = \mu(\rho(0)) = \mu(1) = n - 1$$

Since the only rotation sending 0 to $n - 1$ is ρ^{n-1} , we get $\mu\rho\mu = \rho^{n-1}$. Multiplying both sides on the left by μ (and using $\mu^2 = \iota$):

$$\rho\mu = \mu\rho^{n-1}$$

The general relation $\rho^k\mu = \mu\rho^{n-k}$ follows by induction: assuming $\rho^{k-1}\mu = \mu\rho^{n-(k-1)}$,

$$\rho^k\mu = \rho(\rho^{k-1}\mu) = \rho\mu\rho^{n-k+1} = (\mu\rho^{n-1})\rho^{n-k+1} = \mu\rho^{2n-k} = \mu\rho^{n-k}$$

where the last step uses $\rho^n = \iota$. □

Note:

The relation $\rho^k\mu = \mu\rho^{n-k}$ is the tool for rewriting any product in D_n into standard form. It says: to “move μ past ρ^k ,” replace the exponent k with $n - k$.

Arithmetic in D_n

The following facts are all we need for computations:

1. $\rho^n = \iota$ (rotation by 2π is the identity)
2. $(\rho^k)^{-1} = \rho^{n-k}$
3. $\mu^2 = \iota$, so $\mu^{-1} = \mu$ (reflecting twice gives the identity)
4. $\rho^k\mu = \mu\rho^{n-k}$ (use this to move all μ 's to the left)

Example 1.4.8: Compute $(\mu\rho^2)(\mu\rho)$ in D_5

$$\begin{aligned} (\mu\rho^2)(\mu\rho) &= \mu\rho^2\mu\rho \\ &= \mu(\rho^2\mu)\rho \\ &= \mu(\mu\rho^{5-2})\rho \quad (\text{using rule 4 with } k = 2, n = 5) \\ &= \mu^2\rho^3\rho \\ &= \iota \cdot \rho^4 \\ &= \rho^4 \end{aligned}$$

Example 1.4.9: Compute $(\mu\rho^k)^{-1}$ in D_n

$$\begin{aligned}
(\mu\rho^k)^{-1} &= (\rho^k)^{-1}\mu^{-1} \\
&= \rho^{n-k}\mu \\
&= \mu\rho^{n-(n-k)} \quad (\text{using rule 4}) \\
&= \mu\rho^k
\end{aligned}$$

So every reflection $\mu\rho^k$ is its own inverse. This makes geometric sense: reflecting across a line twice returns every point to its original position.

Example 1.4.10: Compute $\rho\mu\rho^{-1}$ in D_{12}

$$\begin{aligned}
\rho\mu\rho^{-1} &= \rho\mu\rho^{11} \\
&= (\rho\mu)\rho^{11} \\
&= (\mu\rho^{11})\rho^{11} \\
&= \mu\rho^{22} \\
&= \mu\rho^{10} \quad (\text{since } \rho^{12} = \iota)
\end{aligned}$$

Theorem 1.4.7 (D_n is nonabelian for $n \geq 3$): For $n \geq 3$, the dihedral group D_n is nonabelian.

Proof:

We check whether $\mu\rho = \rho\mu$ by evaluating both at 0:

$$\begin{aligned}
\mu\rho(0) &= \mu(1) = n - 1 \\
\rho\mu(0) &= \rho(0) = 1
\end{aligned}$$

Since $n \geq 3$, we have $n - 1 \neq 1$, so $\mu\rho \neq \rho\mu$. □

Note:-

For $n = 3$, we have $D_3 \cong S_3$, the symmetric group on 3 letters. Both have 6 elements, and the isomorphism sends $\rho \mapsto (0, 1, 2)$ and $\mu \mapsto (1, 2)$. Since S_3 is the smallest nonabelian group, and $D_3 \cong S_3$, the dihedral group D_3 is the smallest nonabelian group.

1.5 Subgroups

Definition 1.5.1: Subgroup

A subset H of a group G is called a **subgroup** of G if H is itself a group under the induced operation from G . We write $H \leq G$ to denote that H is a subgroup of G .

Definition 1.5.2: Improper and Trivial Subgroups

Every group G has two obvious subgroups: G itself and $\{e\}$, where e is the identity element. The subgroup G is called the **improper subgroup**, and all other subgroups are **proper subgroups**. The subgroup $\{e\}$ is the **trivial subgroup**, and all other subgroups are **nontrivial**.

Example 1.5.1

We have the chain of subgroups $\mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$ under addition. Each is a proper subgroup of the next.

The n^{th} roots of unity U_n form a subgroup of U (the unit circle group), which in turn is a subgroup of $\mathbb{C}^*$ (the nonzero complex numbers under multiplication). That is, $U_n \leq U \leq \mathbb{C}^*$.

For any $n \geq 3$, the dihedral group D_n is a subgroup of $S_{\mathbb{Z}_n}$, the symmetric group on $\mathbb{Z}_n$, since every symmetry of the regular n -gon is in particular a permutation of its vertices. That is, $D_n \leq S_{\mathbb{Z}_n}$.

Note:-

If $H \leq G$ and $a \in H$, then the identity element of H must be the same as the identity element e of G . This is because the equation $ax = a$ has a unique solution in G (namely e), and since H is itself a group, the solution in H must be that same e . A similar argument shows that the inverse of a in H is the same as a^{-1} in G .

Example 1.5.2

The group $\mathbb{Z}_4$ has only one nontrivial proper subgroup: $\{0, 2\}$. Note that $\{0, 3\}$ is *not* a subgroup since it is not closed under $+_4$: $3 +_4 3 = 2 \notin \{0, 3\}$.

The Klein 4-group $V = \{e, a, b, c\}$ has three nontrivial proper subgroups: $\{e, a\}$, $\{e, b\}$, and $\{e, c\}$. Note that $\{e, a, b\}$ is not a subgroup since $ab = c \notin \{e, a, b\}$.

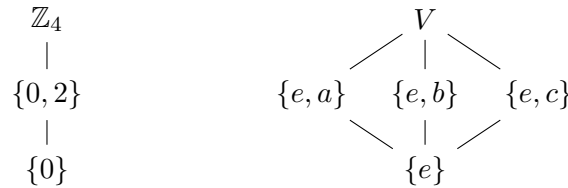


Figure 1.5.1: Subgroup diagram for $\mathbb{Z}_4$ (left), Subgroup diagram for V (right)

1.5.1 Subgroup Tests

Theorem 1.5.1 (Two-Step Subgroup Test): A subset H of a group G is a subgroup of G if and only if:

1. H is closed under the binary operation of G :

$$a, b \in H \implies ab \in H$$

2. For all $a \in H$, $a^{-1} \in H$:

$$a \in H \implies a^{-1} \in H$$

Proof:

($\Rightarrow$) If H is a subgroup, then by definition it must satisfy the group properties. Therefore, for any $a, b \in H$, the product ab must also be in H (closure), and for any $a \in H$, its inverse a^{-1} must also be in H (inverses).

($\Leftarrow$) If H satisfies the two conditions, then it is closed under the operation and contains inverses. Since H is nonempty, it contains an identity element (the identity of G). Associativity is inherited from G . Therefore, H is a subgroup of G . $\square$

Note:-

Since $a \in H \implies a^{-1} \in H$, the identity element e of G must be in H . Because, if $a \in H$ and $a^{-1} \in H$, then by closure $aa^{-1} = e \in H$.

Example 1.5.3

Let G be the multiplicative group of all invertible $n \times n$ complex matrices, and let $T = \{A \in G \mid \det(A) = 1\}$. We verify $T \leq G$ using the two-step test:

1. **Closure:** If $\det(A) = 1$ and $\det(B) = 1$, then $\det(AB) = \det(A)\det(B) = 1$, so $AB \in T$.
2. **Inverses:** If $\det(A) = 1$, then $\det(A^{-1}) = 1/\det(A) = 1$, so $A^{-1} \in T$.

Hence, T is a subgroup of G .

Theorem 1.5.2 (One-Step Subgroup Test): A nonempty subset H of a group G is a subgroup of G if and only if for all $a, b \in H$, $ab^{-1} \in H$.

Proof:

($\Rightarrow$) If $H \leq G$, then H is a group under the induced operation, so for all $a, b \in H$, we have $b^{-1} \in H$ and $ab^{-1} \in H$.

($\Leftarrow$) If for all $a, b \in H$, we have $ab^{-1} \in H$, then we can verify the conditions of the two-step test:

1. **Closure:** If $a, b \in H$, then $ab^{-1} \in H$ by assumption. Since $b^{-1} \in H$, we can take b^{-1} as the second element to get $a(b^{-1})^{-1} = ab \in H$.
2. **Inverses:** If $a \in H$, then taking $b = e$ gives $ae^{-1} = a \in H$. Taking $a = e$ gives $eb^{-1} = b^{-1} \in H$.

Thus, H satisfies the conditions of the two-step test, so $H \leq G$. □

Theorem 1.5.3 (Finite Subgroup Test): Let H be a finite nonempty subset of a group G . Then H is a subgroup of G if and only if H is closed under the operation of G .

Note:-

This is a very convenient test for finite subsets: we only need to check closure! The identity and inverses come for free. The key idea is that if $a \in H$ and H is finite, then the elements $a, a^2, a^3, \dots$ cannot all be distinct, so $a^i = a^j$ for some $i > j$, giving $a^{i-j} = e \in H$. Then $a^{i-j-1} = a^{-1} \in H$.

Example 1.5.4

We can verify that U_n is a subgroup of $\mathbb{C}^*$ using the finite subgroup test. Since U_n has exactly n elements and is finite, we only need to check closure: if $z_1^n = 1$ and $z_2^n = 1$, then $(z_1 z_2)^n = z_1^n z_2^n = 1$. Hence $U_n \leq \mathbb{C}^*$.

Example 1.5.5

The subset $H = \{1, \rho, \rho^2, \dots, \rho^{n-1}\} \subset D_n$ consisting of all rotations is a subgroup of D_n . By the finite subgroup test, we only need to check closure: for $k, r \in \mathbb{Z}_n$, we have $\rho^k \rho^r = \rho^{k+r} \in H$. Therefore $H \leq D_n$.

Theorem 1.5.4 (Intersection of Subgroups): If H and K are subgroups of a group G , then $H \cap K$ is also a subgroup of G . More generally, the intersection of any nonempty collection of subgroups of G is a subgroup of G .

Proof:

We verify the conditions of the two-step test for $H \cap K$:

1. **Closure:** Let $a, b \in H \cap K$. Then $a, b \in H$ and $a, b \in K$. Since H and K are subgroups, $ab \in H$ and $ab \in K$, so $ab \in H \cap K$.
2. **Inverses:** If $a \in H \cap K$, then $a \in H$ and $a \in K$. Since H and K are subgroups, $a^{-1} \in H$ and $a^{-1} \in K$, so $a^{-1} \in H \cap K$.

The generalization to an arbitrary collection $\{H_i \mid i \in I\}$ of subgroups follows by the same argument, replacing “ H and K ” with “all H_i ”. $\square$

Note:-

The collection of subgroups of G can be denoted as $\bigcap_{i \in I} H_i$, where I is an index set and each H_i is a subgroup of G . The intersection $\bigcap_{i \in I} H_i$ consists of all elements that are in every subgroup H_i . Since the intersection of subgroups is itself a subgroup, $\bigcap_{i \in I} H_i$ is a subgroup of G .

Warning about unions

While the intersection of subgroups is always a subgroup, the *union* of subgroups is generally **not** a subgroup. For example, in $\mathbb{Z}$, both $2\mathbb{Z}$ and $3\mathbb{Z}$ are subgroups, but $2\mathbb{Z} \cup 3\mathbb{Z}$ is not closed under addition since $2 + 3 = 5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$.

1.5.2 Cyclic Subgroups

Theorem 1.5.5: Let G be a group and let $a \in G$. Then

$$H = \langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$$

is a subgroup of G and is the smallest subgroup of G that contains a . That is, every subgroup containing a contains H .

Proof:

We verify the conditions of the two-step subgroup test:

1. **Closure:** For $a^r, a^s \in H$, we have $a^r a^s = a^{r+s} \in H$.
2. **Inverses:** For $a^r \in H$, we have $a^{-r} \in H$ and $a^{-r} a^r = e$.

Hence, $H \leq G$.

To see that H is the smallest such subgroup, observe that any subgroup of G containing a must contain a^n for all $n \in \mathbb{Z}$ (by closure, identity, and inverse axioms), and hence must contain H . $\square$

Definition 1.5.3: Cyclic Subgroup Generated by a

The subgroup $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$ is called the **cyclic subgroup of G generated by a** .

Definition 1.5.4: Generator and Cyclic Group

An element a of a group G is a **generator** of G if $\langle a \rangle = G$. A group G is **cyclic** if there exists some element $a \in G$ such that $G = \langle a \rangle$.

Example 1.5.6

The group $\mathbb{Z}_4$ is cyclic with generators 1 and 3, since $\langle 1 \rangle = \langle 3 \rangle = \mathbb{Z}_4$. However, V is not cyclic, because $\langle a \rangle = \{e, a\}$, $\langle b \rangle = \{e, b\}$, and $\langle c \rangle = \{e, c\}$ are all proper subgroups.

Example 1.5.7

The group $\mathbb{Z}$ under addition is cyclic, generated by 1 (or -1). In $\mathbb{Z}$, the cyclic subgroup generated by n is $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$, which is the set of all multiples of n . For example, $\langle 3 \rangle = 3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\}$.

Example 1.5.8

In D_{10} , consider $\langle \mu\rho^k \rangle$ for any $k \in \mathbb{Z}_{10}$. Since $(\mu\rho^k)^2 = \iota$, we have $\langle \mu\rho^k \rangle = \{\iota, \mu\rho^k\}$. On the other hand, since $\rho^{-1} = \rho^9$ and $\rho^{10} = \iota$, every negative power of ρ is also a positive power, so $\langle \rho \rangle = \{\iota, \rho, \rho^2, \dots, \rho^9\}$, which is the subgroup of all rotations.

Definition 1.5.5: Order/Period of an Element

Let a be an element of a group G . If the cyclic subgroup $\langle a \rangle$ is finite, then the **order (or period) of a** is $|\langle a \rangle|$. Otherwise, a is said to have **infinite order**.

Note:-

We will see shortly that the order of a is the smallest positive integer n such that $a^n = e$, provided such an n exists. If no such n exists, then a has infinite order.

1.6 Cyclic Groups

Cyclic groups are fundamental building blocks in group theory. Every finite Abelian group can be decomposed into cyclic groups, so understanding their structure is essential.

Definition 1.6.1: Cyclic Group

A group G is called a **cyclic group** if there exists an element $g \in G$ such that every element in G can be expressed as g^n for some integer n .

$$G = \langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$$

The element g is called a **generator** of the group G .

1.6.1 Elementary Properties of Cyclic Groups

Theorem 1.6.1: Every cyclic group is Abelian.

Proof:

Let $G = \langle g \rangle$ be a cyclic group. If $g_1, g_2 \in G$, then $g_1 = g^r$ and $g_2 = g^s$ for some $r, s \in \mathbb{Z}$. Then

$$g_1 g_2 = g^r g^s = g^{r+s} = g^{s+r} = g^s g^r = g_2 g_1$$

so G is Abelian. □

Note:-

The converse is false: not every Abelian group is cyclic. For instance, the Klein 4-group V is Abelian but not cyclic.

1.6.2 The Division Algorithm

Theorem 1.6.2 (Division Algorithm for $\mathbb{Z}$): If m is a positive integer and n is any integer, then there exist unique integers q (quotient) and r (remainder) such that

$$n = mq + r \quad \text{where } 0 \leq r < m$$

Note:-

The idea is simple: on the number line, n either lands on a multiple of m (then $r = 0$), or it falls between two consecutive multiples qm and $(q + 1)m$, and r is the distance from qm to n . This is the same long division taught in elementary school, formalized for all integers.

Example 1.6.1

When 38 is divided by 7: since $35 = 7 \cdot 5$, we get $38 = 7(5) + 3$, so $q = 5$ and $r = 3$.

When -38 is divided by 7: since $-42 = 7(-6)$, we get $-38 = 7(-6) + 4$, so $q = -6$ and $r = 4$. Note that the remainder must be nonnegative.

1.6.3 Subgroups of Cyclic Groups

Theorem 1.6.3: A subgroup of a cyclic group is cyclic.

Proof:

Let $G = \langle a \rangle$ be a cyclic group and let $H \leq G$. If $H = \{e\}$, then $H = \langle e \rangle$ is cyclic. If $H \neq \{e\}$, then $a^n \in H$ for some $n \in \mathbb{Z}^+$ (if $a^{-n} \in H$, then $(a^{-n})^{-1} = a^n \in H$ too). Let m be the smallest positive integer such that $a^m \in H$. We claim $H = \langle a^m \rangle$.

Let $b \in H$. Since $b \in G = \langle a \rangle$, we have $b = a^n$ for some $n \in \mathbb{Z}$. By the division algorithm, $n = mq + r$ where $0 \leq r < m$. Then

$$a^r = a^{n-mq} = a^n(a^m)^{-q}$$

Since $a^n \in H$ and $a^m \in H$ (hence $(a^m)^{-q} \in H$), we get $a^r \in H$. But m was the smallest positive integer with $a^m \in H$ and $0 \leq r < m$, so $r = 0$. Thus $n = mq$ and $b = a^n = (a^m)^q$, showing $b \in \langle a^m \rangle$. □

Corollary 1.6.4: The subgroups of $\mathbb{Z}$ under addition are precisely the groups $n\mathbb{Z}$ for $n \in \mathbb{Z}_{\geq 0}$, i.e., every subgroup of $\mathbb{Z}$ is of the form $\{nk \mid k \in \mathbb{Z}\}$ for some nonnegative integer n .

1.6.4 Greatest Common Divisor via Subgroups

Definition 1.6.2: Greatest Common Divisor

Let r be a positive integer and s a nonneg. integer. The set

$$H = \{nr + ms \mid n, m \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{Z}$ under addition (can be verified using the one-step test). Since every subgroup of $\mathbb{Z}$ is cyclic, $H = d\mathbb{Z}$ for some positive integer d . This d is the **greatest common divisor** of r and s , written $d = \gcd(r, s)$.

Bézout's Identity

Since $d \in H$, we can write $d = nr + ms$ for some integers n, m . This is called **Bézout's identity**.

Also, since $r = 1 \cdot r + 0 \cdot s \in H = d\mathbb{Z}$ and $s = 0 \cdot r + 1 \cdot s \in H = d\mathbb{Z}$, we see that d divides both r and s . Any common divisor of r and s must divide $nr + ms = d$, so d is indeed the greatest such divisor.

Note:-

Two positive integers r and s are **relatively prime** if $\gcd(r, s) = 1$.

A useful fact: if r and s are relatively prime and $r \mid sm$, then $r \mid m$. This is because $1 = ar + bs$ for some $a, b \in \mathbb{Z}$, so $m = arm + bsm$, and r divides both terms on the right.

1.6.5 The Structure of Cyclic Groups

Theorem 1.6.5 (Classification of Cyclic Groups): Let $G = \langle a \rangle$ be a cyclic group.

- If G has infinite order, then $G \cong \langle \mathbb{Z}, + \rangle$.
- If G has finite order n , then $G \cong \langle \mathbb{Z}_n, +_n \rangle$.

Proof:

Case 1 (infinite order): Define $\phi : \mathbb{Z} \rightarrow G$ by $\phi(n) = a^n$. We verify that ϕ is an isomorphism:

- **Homomorphism:** For $m, n \in \mathbb{Z}$, we have $\phi(m + n) = a^{m+n} = a^m a^n = \phi(m)\phi(n)$.
- **Injective:** If $\phi(n) = e$, then $a^n = e$. Since G has infinite order, the only power of a that equals e is $a^0 = e$, so $n = 0$. Thus $\ker(\phi) = \{0\}$ and ϕ is injective.
- **Surjective:** For any $g \in G$, we have $g = a^k$ for some $k \in \mathbb{Z}$, so $\phi(k) = g$. Thus ϕ is surjective.

Case 2 (finite order): Let $n = |G|$. Define $\phi : \mathbb{Z}_n \rightarrow G$ by $\phi(k) = a^k$. We verify that ϕ is an isomorphism:

- **Homomorphism:** For $k, r \in \mathbb{Z}_n$, we have $\phi(k +_n r) = a^{k+_n r} = a^k a^r = \phi(k)\phi(r)$.
- **Injective:** If $\phi(k) = e$, then $a^k = e$. Since G has order n , the smallest positive integer m such that $a^m = e$ is n . Thus, if $0 < k < n$, then $a^k \neq e$. The only way for a^k to equal e is if $k = 0$ in $\mathbb{Z}_n$. Hence $\ker(\phi) = \{0\}$ and ϕ is injective.
- **Surjective:** For any $g \in G$, we have $g = a^k$ for some $k \in \mathbb{Z}_n$, so $\phi(k) = g$. Thus ϕ is surjective.

□

Note:-

This theorem tells us there is essentially only one cyclic group of each order (up to isomorphism): $\mathbb{Z}$ for infinite order, and $\mathbb{Z}_n$ for order n . We can visualize the elements of a finite cyclic group $\langle a \rangle$ of order n as points evenly spaced around a circle, with multiplication corresponding to “adding angles” modulo n :

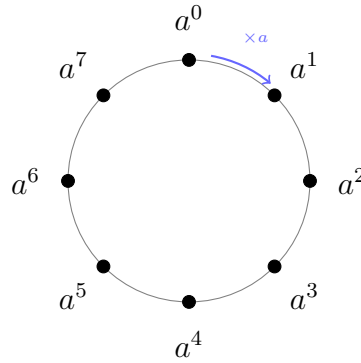


Figure 1.6.1: Elements of a cyclic group of order 8 on a circle

Corollary 1.6.6: *The order of an element a in a group G is the smallest positive integer n such that $a^n = e$, if such n exists. Otherwise, a has infinite order.*

Example 1.6.2: Order of a k -cycle

The order of a k -cycle $\sigma = (a_1, a_2, \dots, a_k)$ in S_n is k . Applying σ repeatedly cycles each element through the k positions, and after exactly k applications, every element returns to its original position.

Example 1.6.3

In $\mathbb{Z}_{12}$ (additive notation), the order of 3 is $12/\gcd(12, 3) = 12/3 = 4$, and $\langle 3 \rangle = \{0, 3, 6, 9\}$. The order of 5 is $12/\gcd(12, 5) = 12/1 = 12$, so 5 generates all of $\mathbb{Z}_{12}$.

1.6.6 Subgroups of Finite Cyclic Groups

Theorem 1.6.7 (Subgroups of $\mathbb{Z}_n$): *Let $G = \langle a \rangle$ be a cyclic group of order n , and let $b = a^s \in G$. Then b generates a cyclic subgroup H of G containing n/d elements, where $d = \gcd(n, s)$. Moreover, $\langle a^s \rangle = \langle a^t \rangle$ if and only if $\gcd(n, s) = \gcd(n, t)$.*

Proof:

By the classification theorem, H has as many elements as the smallest positive integer m such that $b^m = e$. Now $b^m = (a^s)^m = a^{sm}$, and $a^{sm} = e$ if and only if $n \mid sm$.

Let $d = \gcd(n, s)$. We want the smallest m such that $n \mid sm$, i.e., such that $\frac{sm}{n} = \frac{(s/d)m}{n/d}$ is an integer. Since $\gcd(s/d, n/d) = 1$ (dividing out the gcd makes them coprime), we need $\frac{n}{d} \mid m$. Thus the smallest such m is n/d , and $|H| = n/d$.

For the second claim: since $\langle a^s \rangle$ has $n/\gcd(n, s)$ elements and there is exactly one subgroup of each order dividing n in a cyclic group, $\langle a^s \rangle = \langle a^t \rangle$ iff $\gcd(n, s) = \gcd(n, t)$. □

Corollary 1.6.8 (Generators of $\mathbb{Z}_n$): If a is a generator of a finite cyclic group G of order n , then a^r is also a generator of G if and only if $\gcd(r, n) = 1$, i.e., r is relatively prime to n .

Example 1.6.4

To find the generators of $\mathbb{Z}_{18}$: we need $\gcd(r, 18) = 1$. The values of r with $0 \leq r < 18$ that are relatively prime to 18 are 1, 5, 7, 11, 13, 17. So these are exactly the generators of $\mathbb{Z}_{18}$.

Example 1.6.5: All subgroups of $\mathbb{Z}_{18}$

Since $\mathbb{Z}_{18}$ is cyclic, every subgroup is cyclic. For each divisor d of 18, there is exactly one subgroup of order d , generated by $18/d$:

- $\langle 1 \rangle = \mathbb{Z}_{18}$ (order 18)
- $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14, 16\}$ (order 9)
- $\langle 3 \rangle = \{0, 3, 6, 9, 12, 15\}$ (order 6)
- $\langle 6 \rangle = \{0, 6, 12\}$ (order 3)
- $\langle 9 \rangle = \{0, 9\}$ (order 2)
- $\langle 0 \rangle = \{0\}$ (order 1)

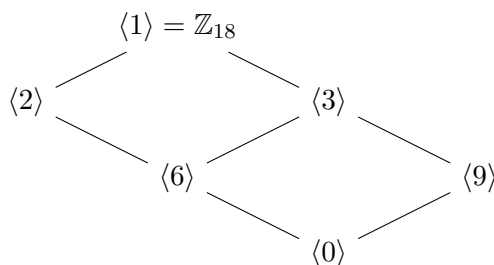


Figure 1.6.2: Subgroup diagram for $\mathbb{Z}_{18}$

Corollary 1.6.9: If G is a finite cyclic group of order n and $H \leq G$, then $|H|$ divides $|G|$.

Proof:

Since G is cyclic, H is cyclic by the earlier theorem. If $G = \langle a \rangle$ and H is generated by a^s , then $|H| = n / \gcd(n, s)$, which divides $n = |G|$. $\square$

Note:-

This result foreshadows Lagrange's Theorem, which states that $|H|$ divides $|G|$ for *any* finite group G and subgroup H , not just cyclic ones. We will see this in a later section.

1.7 Generating Sets and Cayley Digraphs

1.7.1 Generating Sets

So far, we have talked about cyclic groups, which are generated by a single element. But many groups require more than one generator. The idea extends naturally: given elements $a_1, a_2, \dots \in G$, the subgroup generated by them consists of all finite products of their integer powers.

Definition 1.7.1: Intersection of sets

Let $\{S_i \mid i \in I\}$ be a collection of sets. Here, I may be any set of indices. The **intersection** $\bigcap_{i \in I} S_i$ of the sets S_i is the set of all elements that are in all the sets S_i . That is,

$$\bigcap_{i \in I} S_i = \{x \mid \forall i \in I, x \in S_i\}$$

If I is finite, $I = \{1, 2, \dots, n\}$ we may denote $\bigcap_{i \in I} S_i$ by

$$S_1 \cap S_2 \cap \dots \cap S_n$$

Theorem 1.7.1: For any group G and any nonempty collection of subgroups $\{H_i \leq G \mid i \in I\}$, the intersection of all the subgroups H_i , $\bigcap_{i \in I} H_i$, is also a subgroup of G .

Proof:

We verify the conditions of the two-step test for $\bigcap_{i \in I} H_i$:

1. **Closure:** Let $a, b \in \bigcap_{i \in I} H_i$. Then $a, b \in H_i$ for every $i \in I$. Since each H_i is a subgroup, $ab \in H_i$ for every i , so $ab \in \bigcap_{i \in I} H_i$.
2. **Inverses:** If $a \in \bigcap_{i \in I} H_i$, then $a \in H_i$ for every i . Since each H_i is a subgroup, $a^{-1} \in H_i$ for every i , so $a^{-1} \in \bigcap_{i \in I} H_i$.

Thus, $\bigcap_{i \in I} H_i$ is a subgroup of G . □

Definition 1.7.2: Subgroup Generated by a Set

Let G be a group and let $\{a_i \mid i \in I\} \subseteq G$. The **subgroup generated by** $\{a_i \mid i \in I\}$ is the smallest subgroup of G containing all the a_i . If this subgroup equals G itself, then $\{a_i \mid i \in I\}$ **generates** G , and the a_i are called **generators of** G . A group is **finitely generated** if it has a finite generating set.

Note:-

The “smallest subgroup containing all the a_i ” exists because the intersection of any collection of subgroups is again a subgroup. So we can take the intersection of all subgroups of G that contain all the a_i .

Theorem 1.7.2: If G is a group and $\{a_i \mid i \in I\}$ is a nonempty subset of G , then the subgroup H generated by $\{a_i\}$ consists of precisely those elements of G that can be written as finite products of integer powers of the a_i (where powers of a fixed a_i may occur multiple times in the product).

Proof:

Let H be the subgroup generated by $\{a_i \mid i \in I\}$, which is the smallest subgroup of G containing every a_i . Now, define the set:

$$K = \{a_{i_1}^{n_1} \cdot a_{i_2}^{n_2} \cdots a_{i_k}^{n_k} \mid k \in \mathbb{Z}^+, i_j \in I, n_j \in \mathbb{Z}\} \cup \{e\}$$

We want to show that $H = K$.

- **K is a subgroup of G :** First, we verify that K is a subgroup of G :

1. **Closure:** Let $x = a_{i_1}^{n_1} \cdots a_{i_k}^{n_k}$ and $y = a_{j_1}^{m_1} \cdots a_{j_l}^{m_l}$ be elements of K . Then $xy = a_{i_1}^{n_1} \cdots a_{i_k}^{n_k} a_{j_1}^{m_1} \cdots a_{j_l}^{m_l}$, which is also an element of K since it is a finite product of integer powers of the a_i .
2. **Inverses:** Let $x = a_{i_1}^{n_1} \cdots a_{i_k}^{n_k}$ be an element of K . Then $x^{-1} = a_{i_k}^{-n_k} \cdots a_{i_1}^{-n_1}$, which is also an element of K since it is a finite product of integer powers of the a_i .

Thus, $K \leq G$.

Each $a_i = a_i^1$ is a finite product of length 1 with integer exponent, so

$$\forall i \in I, a_i \in K$$

- **K contains every a_i :** Each $a_i = a_i^1$ is a finite product of length 1 with integer exponent, so $a_i \in K$ for every $i \in I$.
- **$H \subseteq K$:** Since H is the smallest subgroup containing every a_i , and K is a subgroup containing every a_i , we must have $H \subseteq K$.
- **$K \subseteq H$:** Since H is a subgroup containing every a_i , it must contain every integer power of each a_i (by closure and inverses). Since H is closed under the group operation, it must contain every finite product of integer powers of the a_i . Thus, $K \subseteq H$.

□

Example 1.7.1

The dihedral group D_n is generated by $\{\mu, \rho\}$: every element can be written as ρ^k or $\mu\rho^k$ for $0 \leq k < n$. It is also generated by $\{\mu, \mu\rho\}$ since $\rho = \mu(\mu\rho)$. Interestingly, both μ and $\mu\rho$ have order 2, while in the set $\{\mu, \rho\}$, one generator has order 2 and the other has order n .

Example 1.7.2

The Klein 4-group $V = \{e, a, b, c\}$ is generated by $\{a, b\}$ since $ab = c$. It is also generated by $\{a, c\}$ or $\{b, c\}$. Adding more elements to a generating set still generates the group: $\{a, b, c\}$ generates V too.

Example 1.7.3

The group $\mathbb{Z}_6$ is generated by $\{1\}$ (cyclic!) and by $\{5\}$. It is also generated by $\{2, 3\}$ since $2 + 3 = 5$, and 5 generates $\mathbb{Z}_6$. However, $\{2, 4\}$ does *not* generate $\mathbb{Z}_6$ since $\langle 2 \rangle = \{0, 2, 4\}$ and $\langle 4 \rangle = \{0, 4, 2\}$, so $\langle 2, 4 \rangle = \{0, 2, 4\}$ is a proper subgroup of $\mathbb{Z}_6$.

1.7.2 Cayley Digraphs

For each generating set S of a finite group G , we can draw a **Cayley digraph** (directed graph) that visually represents the group structure:

- One **vertex** (dot) for each element of G .
- For each generator $s \in S$, a type of **arc** (arrow). An arc of type s from vertex g to vertex h means $gs = h$.
- If a generator s satisfies $s^2 = e$ (it is its own inverse), we draw the arc without an arrow-head.

Properties of Cayley Digraphs

Every Cayley digraph for a group satisfies:

1. **Connected:** We can get from any vertex to any other by traveling along arcs (because $gx = h$ always has a solution).
2. **At most one arc from g to h :** The solution of $gx = h$ is unique.
3. **Regularity:** Each vertex has exactly one outgoing and one incoming arc of each type (for generator b , gb is unique, and $xb = g$ has unique solution $x = gb^{-1}$).
4. **Consistency:** If two different sequences of arc types starting from g lead to h , then those same sequences starting from any vertex u lead to the same vertex v .

Conversely, any digraph satisfying these four properties is a Cayley digraph for some group.

Example 1.7.4: Cayley digraph for $\mathbb{Z}_6$ with $S = \{1\}$

With a single generator, we get a directed cycle through all elements:

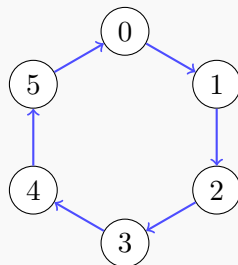
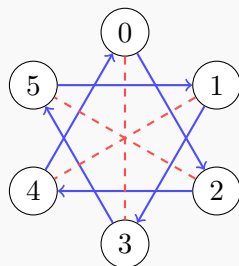


Figure 1.7.1: Cayley digraph for $\mathbb{Z}_6$, $S = \{1\}$

Example 1.7.5: Cayley digraph for $\mathbb{Z}_6$ with $S = \{2, 3\}$

With two generators, we use solid arrows for 2 and dashed (arrowless) edges for 3 (since $3 + 3 = 0$ in $\mathbb{Z}_6$, so 3 is its own inverse):

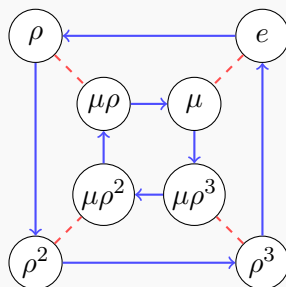


Solid (blue) = generator 2, Dashed (red) = generator 3

Figure 1.7.2: Cayley digraph for $\mathbb{Z}_6$, $S = \{2, 3\}$

Example 1.7.6: Cayley digraph for D_4 with $S = \{\rho, \mu\}$

The dihedral group D_4 has 8 elements. Using ρ (rotation by $\pi/2$) as solid arrows and μ (a reflection, order 2) as dashed edges:



Solid (blue) = ρ , Dashed (red) = μ

Figure 1.7.3: Cayley digraph for D_4 , $S = \{\rho, \mu\}$

Note:-

From the Cayley digraph of D_4 , we can read off products directly. For example, to compute $\mu\rho^2 \cdot \mu$: start at $\mu\rho^2$, follow a dashed edge to arrive at ρ^2 . So $\mu\rho^2 \cdot \mu = \rho^2$.

We can also detect whether a group is Abelian from its Cayley digraph: if for every pair of generators, following generator a then b from any vertex always gives the same result as following b then a , the group is Abelian. For D_4 , this fails: from e , ρ then μ gives $\mu\rho^3$, but μ then ρ gives $\mu\rho$.

Note:-

A Cayley digraph makes the group's structure visible. The outer cycle in the D_4 diagram shows the cyclic subgroup of rotations $\langle \rho \rangle$, and the dashed edges connect each rotation to its corresponding reflection. The fact that the inner cycle runs in the *opposite* direction reflects the relation $\rho\mu = \mu\rho^{-1}$.

2 Structure of Groups

2.1 Groups of Permutations

Before diving deeper into permutations, we introduce a fundamental concept that relates different groups by preserving their mathematical structure.

2.1.1 Homomorphisms

Recall that an isomorphism between two groups is a bijection that preserves the group operation. A homomorphism is a more general mapping that preserves the group operation but is not necessarily one-to-one or onto.

Definition 2.1.1: Homomorphism

Let G and G' be groups with a function $\phi : G \rightarrow G'$. The map ϕ is a **homomorphism** if the homomorphism property

$$\phi(ab) = \phi(a)\phi(b)$$

holds for all $a, b \in G$.

Here, the left-hand side uses the group operation in G , while the right-hand side uses the group operation in G' . If ϕ is a homomorphism that is also a bijection, then ϕ is an isomorphism.

Example 2.1.1

Let $\phi : \mathbb{R} \rightarrow U$ (the circle group) be defined by $\phi(x) = e^{2\pi i x}$. Since $e^{2\pi i(a+b)} = e^{2\pi i a} e^{2\pi i b}$, we have $\phi(a+b) = \phi(a)\phi(b)$. Thus, ϕ is a homomorphism. It maps $\mathbb{R}$ onto U , but is not one-to-one (e.g., $\phi(1) = \phi(0) = 1$), so it is not an isomorphism.

Definition 2.1.2: Image and Inverse Image

If $\phi : X \rightarrow Y$ is a function, $A \subseteq X$, and $B \subseteq Y$, we define the image of A as $\phi[A] = \{\phi(a) \mid a \in A\}$ and the inverse image of B as $\phi^{-1}[B] = \{a \in A \mid \phi(a) \in B\}$.

Theorem 2.1.1 (Properties of Homomorphisms): Let ϕ be a homomorphism of a group G into a group G' .

1. If e is the identity in G , then $\phi(e)$ is the identity e' in G' .
2. If $a \in G$, then $\phi(a^{-1}) = \phi(a)^{-1}$.
3. If H is a subgroup of G , then $\phi[H]$ is a subgroup of G' .
4. If K' is a subgroup of G' , then $\phi^{-1}[K']$ is a subgroup of G .

Loosely speaking, homomorphisms preserve the identity, inverses, and subgroups.

Proof:

1. $\phi(e) = \phi(ee) = \phi(e)\phi(e)$, so $\phi(e)$ is the identity in G' .
2. $\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e'$, so $\phi(a^{-1})$ is the inverse of $\phi(a)$.
3. If $x, y \in \phi[H]$, then $x = \phi(h_1)$ and $y = \phi(h_2)$ for some $h_1, h_2 \in H$. Since H is a subgroup, $h_1 h_2^{-1} \in H$. Then $xy^{-1} = \phi(h_1)\phi(h_2)^{-1} = \phi(h_1)\phi(h_2^{-1}) = \phi(h_1 h_2^{-1})$, which is in $\phi[H]$. Thus, $\phi[H]$ is a subgroup of G' .
4. If $x, y \in \phi^{-1}[K']$, then $\phi(x), \phi(y) \in K'$. Since K' is a subgroup, $\phi(x)\phi(y)^{-1} = \phi(x)\phi(y^{-1}) = \phi(xy^{-1})$ is in K' . Thus, xy^{-1} is in $\phi^{-1}[K']$, and $\phi^{-1}[K']$ is a subgroup of G .

□

Definition 2.1.3: Kernel

Let $\phi : G \rightarrow G'$ be a homomorphism of groups. The subgroup $\phi^{-1}[\{e'\}] = \{x \in G \mid \phi(x) = e'\}$ is called the **kernel** of ϕ , denoted by $\ker(\phi)$.

2.1.2 Cayley's Theorem

Every group is structurally related to a group of permutations. The connection relies on the group table: each row represents a permutation of the group elements because multiplying the entire group by a fixed element x merely permutes the elements.

Definition 2.1.4: Left Regular Representation

Let G be a group. The function $\phi : G \rightarrow S_G$ given by $\phi(x) = \lambda_x$, where $\lambda_x(g) = xg$ for all $g \in G$, is called the **left regular representation** of G .

Theorem 2.1.2 (Cayley's Theorem): Every group is isomorphic to a group of permutations.

Proof:

Let G be a group and let $\phi : G \rightarrow S_G$ be the left regular representation defined by $\phi(x) = \lambda_x$. We must show that ϕ is a one-to-one homomorphism; then $\phi[G]$ is a subgroup of S_G , making G isomorphic to the permutation group $\phi[G]$.

First, suppose $\phi(a) = \phi(b)$. Then $\lambda_a = \lambda_b$, so $\lambda_a(e) = \lambda_b(e)$. This means $ae = be$, so $a = b$. Thus, ϕ is one-to-one.

Next, we show ϕ is a homomorphism: we want $\phi(ab) = \phi(a)\phi(b)$, or $\lambda_{ab} = \lambda_a\lambda_b$. For any $g \in G$:

$$\lambda_{ab}(g) = (ab)g = a(bg) = \lambda_a(bg) = \lambda_a(\lambda_b(g)) = (\lambda_a \circ \lambda_b)(g)$$

Thus ϕ is a homomorphism, and the theorem is proven. $\square$

Definition 2.1.5: Right Regular Representation

Let G be a group. The map $\tau : G \rightarrow S_G$ given by $\tau(x) = \sigma_{x^{-1}}$ where $\sigma(x) = gx$ is called the **right regular representation** of G .

2.1.3 Even and Odd Permutations

Definition 2.1.6: Transposition

A cycle of length 2 is called a **transposition**. A transposition leaves all elements except two fixed and swaps those two.

A simple computation shows that any cycle of length n can be decomposed into transpositions:

$$(a_1, a_2, \dots, a_n) = (a_1, a_n)(a_1, a_{n-1}) \cdots (a_1, a_3)(a_1, a_2)$$

Because every permutation of a finite set is a product of disjoint cycles, we establish:

Theorem 2.1.3: Any permutation of a finite set containing at least two elements can be written as a product of transpositions.

Note:-

Any permutation of length n can be written as a product of at most $n - 1$ transpositions, since a cycle of length n can be decomposed into $n - 1$ transpositions.

Note:-

This decomposition is not unique. For instance, the identity can be written as $(1, 2)(1, 2)$ or $(1, 3)(1, 3)(2, 4)(2, 4)$. A given permutation can therefore be written as the product of different numbers of transpositions. However, the *parity* of the number of transpositions is an invariant.

Definition 2.1.7: Orbit of a Permutation

Let σ be a permutation of a set A . The **orbit** of an element $a \in A$ under σ is the set $\{\sigma^n(a) \mid n \in \mathbb{Z}\}$.

Example 2.1.2

Let $\sigma = (1, 2, 6)(3, 5) \in S_6$. Then the orbit of 1, 2, and 6 is $\{1, 2, 6\}$, the orbit of 3 is $\{3, 5\}$, and the orbit of 4 is $\{4\}$ since 4 is fixed by σ .

Theorem 2.1.4: *No permutation in the symmetric group S_n can be expressed both as a product of an even number of transpositions and as a product of an odd number of transpositions.*

Proof:

Let σ be a permutation of S_n . We will show that the parity of the number of transpositions in any decomposition of σ is invariant.

Consider the action of σ on the set $\{1, 2, \dots, n\}$. The number of orbits under this action is equal to the number of cycles in the cycle decomposition of σ . Each cycle of length k can be written as a product of $k - 1$ transpositions. Therefore, if σ has c cycles, it can be expressed as a product of $n - c$ transpositions.

Since $n - c$ is fixed for a given σ , the parity of the number of transpositions in any decomposition of σ is also fixed. Thus, if σ can be written as a product of an even number of transpositions, it cannot be written as a product of an odd number of transpositions, and vice versa. $\square$

Definition 2.1.8: Even and Odd Permutations

A permutation is **even** if it can be written as a product of an even number of transpositions, and **odd** if it can be written as a product of an odd number of transpositions.

Example 2.1.3

In S_3 , the transpositions are $(1, 2)$, $(1, 3)$, and $(2, 3)$, which are odd. The 3-cycles $(1, 2, 3)$ and $(1, 3, 2)$ can be written as products of two transpositions (e.g., $(1, 2, 3) = (1, 3)(1, 2)$), so they are even. The identity is also even since it can be written as a product of zero transpositions. Thus, S_3 has three odd permutations and three even permutations.

2.1.4 Alternating Groups

Theorem 2.1.5: If $n \geq 2$, then the collection of all even permutations of $\{1, 2, 3, \dots, n\}$ forms a subgroup of order $n!/2$ of the symmetric group S_n .

Proof:

Let A_n be the set of all even permutations in S_n . We will show that A_n is a subgroup of S_n and that its order is $n!/2$.

First, A_n is closed under the group operation since the product of two even permutations is even. Second, if σ is an even permutation, then its inverse σ^{-1} is also even, because the inverse of a product of transpositions is the product of the inverses of those transpositions, and the number of transpositions in the inverse is the same as in the original permutation. Thus, A_n contains inverses. So, A_n is a subgroup of S_n .

Finally, we need to show that A_n is exactly half of S_n . For that, we can define a function called the **sign of a permutation**, $\text{sgn} : S_n \rightarrow \{1, -1\}$ such that

$$\text{sgn}(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is even} \\ -1, & \text{if } \sigma \text{ is odd} \end{cases}$$

Thinking of $\{1, -1\}$ as a group under multiplication, sgn is a homomorphism. Since 1 is the identity in $\{1, -1\}$, $\ker(\text{sgn}) = \text{sgn}^{-1}[\{1\}]$ is a subgroup of S_n that consists of all even permutations, which is exactly A_n . By the First Isomorphism Theorem, $S_n/A_n \cong \{1, -1\}$, so the index of A_n in S_n is 2. Therefore, the order of A_n is $n!/2$. $\square$

Note:-

The set of all odd permutations has order $\frac{n!}{2}$ as well, but it does not form a subgroup since it is not closed under multiplication (the product of two odd permutations is an even permutation).

Definition 2.1.9: Alternating Group

The subgroup of S_n consisting of all even permutations on n letters is the **alternating group** A_n on n letters.

Note:-

Both S_n and A_n are among the most important groups in mathematics, for two deep reasons.

Cayley's Theorem. Every finite group G is isomorphic to a subgroup of S_n for $n = |G|$. The idea is that each element $g \in G$ induces a permutation of G itself via left multiplication ($x \mapsto gx$), and collecting all such permutations yields a subgroup of S_n isomorphic to G . In this sense, S_n is the *universal home* for all finite groups of order $\leq n$ – permutation groups are not merely examples of groups, they are, up to isomorphism, *all* groups.

Insolubility of the quintic. While polynomial equations of degree ≤ 4 admit closed-form solutions in terms of radicals, no such formula exists for degree $n \geq 5$. This is explained by Galois theory: a polynomial is solvable by radicals if and only if its Galois group is *solvable* (i.e., built up from abelian pieces via a subnormal series). For $n \geq 5$, the alternating group A_n is **simple** – it admits no proper normal subgroups – and is non-abelian, making it an indecomposable, unsolvable block. Since A_n embeds into the Galois group of a general degree- n polynomial, the Galois group is not solvable, and no radical formula can exist.

It is remarkable that a purely structural property of A_n – the absence of normal subgroups – turns out to be the precise obstruction to solving polynomial equations by radicals.

2.2 Finitely Generated Abelian Groups

We can use known groups as building blocks to form more groups. A primary way to construct new groups from existing ones is through the direct product.

2.2.1 Direct Products

Definition 2.2.1: Direct Product

Let $G_1, G_2, \dots, G_n$ be groups. The **direct product**

$$\prod_{i=1}^n G_i = G_1 \times G_2 \times \cdots \times G_n$$

is the set of all ordered n -tuples $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$, where $a_i \in G_i$ and $b_i \in G_i$, equipped with the binary operation of multiplication by components:

$$(a_1, a_2, \dots, a_n)(b_1, b_2, \dots, b_n) = (a_1b_1, a_2b_2, \dots, a_nb_n)$$

This makes the Cartesian product a group. The identity is $(e_1, e_2, \dots, e_n)$, and the inverse is $(a_1^{-1}, a_2^{-1}, \dots, a_n^{-1})$.

Proof:

We verify the group axioms:

- **Closure:** If $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$ are in $G_1 \times G_2 \times \cdots \times G_n$, then a_ib_i is in G_i for each i , so their product is in the direct product.
- **Associativity:** The operation is associative because each component operation is associative.
- **Identity:** The tuple of identities $(e_1, e_2, \dots, e_n)$ acts as the identity element.
- **Inverses:** Each element has an inverse given by taking the inverse in each component.

Thus, the direct product is a group. □

Note:-

If the groups are abelian and we use additive notation, the direct product is sometimes called the **direct sum** and denoted $\oplus_{i=1}^n G_i = G_1 \oplus G_2 \oplus \cdots \oplus G_n$. If G_i has r_i elements, the direct product has $r_1r_2 \cdots r_n$ elements.

Example 2.2.1: $\mathbb{Z}_2 \times \mathbb{Z}_3$

Consider $\mathbb{Z}_2 \times \mathbb{Z}_3$, which has dimension $2 \times 3 = 6$. Let's test if it's cyclic by finding the multiples of $(1, 1)$:

$$\begin{aligned} 1(1, 1) &= (1, 1) \\ 2(1, 1) &= (1 +_2 1, 1 +_3 1) = (0, 2) \\ 3(1, 1) &= 2(1, 1) + (1, 1) = (0, 2) + (1, 1) = (1, 0) \\ 4(1, 1) &= 3(1, 1) + (1, 1) = (1, 0) + (1, 1) = (0, 1) \\ 5(1, 1) &= 4(1, 1) + (1, 1) = (0, 1) + (1, 1) = (1, 2) \\ 6(1, 1) &= 5(1, 1) + (1, 1) = (1, 2) + (1, 1) = (0, 0) \end{aligned}$$

Since $(1, 1)$ generates all 6 elements, $\mathbb{Z}_2 \times \mathbb{Z}_3$ is a cyclic group, isomorphic to $\mathbb{Z}_6$.

Theorem 2.2.1: The group $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic and is isomorphic to $\mathbb{Z}_{mn}$ if and only if m and n are relatively prime (i.e., $\gcd(m, n) = 1$).

Proof:

($\Rightarrow$) If $\gcd(m, n) = 1$, then $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$.

The order of $(1, 1)$ is the smallest positive integer k such that $k(1, 1) = (0, 0)$. Since addition is componentwise, this requires:

$$m \mid k \quad \text{and} \quad n \mid k.$$

The smallest such k is $\text{lcm}(m, n)$, so $\text{ord}(1, 1) = \text{lcm}(m, n)$. Since $\gcd(m, n) = 1$, we have:

$$\text{lcm}(m, n) = \frac{mn}{\gcd(m, n)} = mn.$$

So $(1, 1)$ has order $mn = |\mathbb{Z}_m \times \mathbb{Z}_n|$, meaning it generates the entire group. Thus $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic of order mn , and by the Classification Theorem, $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$.

($\Leftarrow$) If $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic, then $\gcd(m, n) = 1$.

We first show that every element of $\mathbb{Z}_m \times \mathbb{Z}_n$ has order dividing $\text{lcm}(m, n)$. Let $L = \text{lcm}(m, n)$. For any (a, b) :

$$L(a, b) = (La \bmod m, Lb \bmod n) = (0, 0),$$

since $m \mid L$ and $n \mid L$. Hence $\text{ord}(a, b)$ divides $L = \text{lcm}(m, n)$ for every (a, b) .

Now suppose $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic. Then some generator has order mn . But every element's order divides $\text{lcm}(m, n)$, so:

$$mn \mid \text{lcm}(m, n).$$

Since $\text{lcm}(m, n) \leq mn$ always, we must have $\text{lcm}(m, n) = mn$, which gives:

$$\frac{mn}{\gcd(m, n)} = mn \implies \gcd(m, n) = 1.$$

This completes the proof. □

Corollary 2.2.2: The group $\prod_{i=1}^n \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 m_2 \dots m_n}$ if and only if the number m_i for $i = 1, \dots, n$ are such that the gcd of any two of them is 1.

Proof:

If $\gcd(m_i, m_j) = 1$ for all $i \neq j$, then we can apply the previous theorem repeatedly to conclude that $\mathbb{Z}_{m_1} \times \mathbb{Z}_{m_2} \times \dots \times \mathbb{Z}_{m_n}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 m_2 \dots m_n}$. Conversely, if the direct product is cyclic, then the order of any element divides the least common multiple of the m_i . Since there exists an element of order $m_1 m_2 \dots m_n$, it must be that $\text{lcm}(m_1, m_2, \dots, m_n) = m_1 m_2 \dots m_n$, which implies that $\gcd(m_i, m_j) = 1$ for all $i \neq j$. □

Definition 2.2.2: Least Common Multiple

Let $r_1, r_2, \dots, r_n$ be positive integers. Their **least common multiple (lcm)** is the smallest positive integer that is a multiple of each r_i for $i = 1, 2, \dots, n$. Formally, it is the positive generator of the cyclic group of all common multiples of the r_i .

Theorem 2.2.3 (Order of an Element in a Direct Product): Let $(a_1, a_2, \dots, a_n) \in \prod_{i=1}^n G_i$. If a_i is of finite order r_i in G_i , then the order of the tuple $(a_1, a_2, \dots, a_n)$ in the direct product is the least common multiple of all the r_i .

Proof:

Let $L = \text{lcm}(r_1, r_2, \dots, r_n)$. We will show that L is the smallest positive integer such that $L(a_1, a_2, \dots, a_n) = (e_1, e_2, \dots, e_n)$.

First, since r_i is the order of a_i , we have $L = r_i \cdot q$ for some positive integer q . So,

$$La_i = r_i \cdot q \cdot a_i = q \cdot (r_i a_i) = q \cdot e_i = e_i$$

This holds for each i , so componentwise, $L(a_1, a_2, \dots, a_n) = (e_1, e_2, \dots, e_n)$.

Now suppose there exists a positive integer $k < L$ such that $k(a_1, a_2, \dots, a_n) = (e_1, e_2, \dots, e_n)$. This implies that $ka_i = e_i$ for each i , which means that r_i divides k for each i . However, this contradicts the definition of L as the least common multiple. Thus, no such k exists and the order of $(a_1, a_2, \dots, a_n)$ is indeed $\text{lcm}(r_1, r_2, \dots, r_n)$. $\square$

Example 2.2.2: Finding Order

Find the order of $(8, 4, 10)$ in $\mathbb{Z}_{12} \times \mathbb{Z}_{60} \times \mathbb{Z}_{24}$.

- The order of 8 in $\mathbb{Z}_{12}$ is $12/\gcd(8, 12) = 12/4 = 3$.
- The order of 4 in $\mathbb{Z}_{60}$ is $60/\gcd(4, 60) = 60/4 = 15$.
- The order of 10 in $\mathbb{Z}_{24}$ is $24/\gcd(10, 24) = 24/2 = 12$.

The lcm of 3, 15, and 12 is 60. Thus, the order of $(8, 4, 10)$ is 60.

We can distinguish between an *external* direct product (formed from independent groups) and an *internal* direct product, where the group is formed by the product of its own subgroups G_i , defined by setting elements equal to identity in all components except the i -th. These structures are fundamentally isomorphic.

2.2.2 The Structure of Finitely Generated Abelian Groups

The following profound theorem classifies all finitely generated abelian groups up to isomorphism. There are two standard ways to present it.

Theorem 2.2.4 (Fundamental Theorem of Finitely Generated Abelian Groups (Primary Factor Version)): Every finitely generated Abelian group G is isomorphic to a direct product of cyclic groups in the form

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$$

where the p_i are primes, not necessarily distinct, and the r_i are positive integers. The direct product is unique except for possible rearrangement of the factors; that is, the number of factors of $\mathbb{Z}$ (called the **Betti number** of G) is unique, and the prime powers $(p_i)^{r_i}$ are unique.

Proof:

(Better look at it later.)

Step 1: Decompose G into torsion and free parts.

Define the torsion subgroup

$$T(G) = \{g \in G : ng = e \text{ for some } n \geq 1\}.$$

Since G is Abelian, $T(G)$ is a subgroup of G . The quotient $G/T(G)$ is torsion-free and finitely generated, hence free, so $G/T(G) \cong \mathbb{Z}^k$ for some $k \geq 0$. This k is called the **Betti number** of

G . Since $\mathbb{Z}^k$ is free, the short exact sequence

$$0 \rightarrow T(G) \rightarrow G \rightarrow \mathbb{Z}^k \rightarrow 0$$

splits, giving $G \cong T(G) \times \mathbb{Z}^k$. It remains to decompose $T(G)$.

Step 2: Decompose $T(G)$ by primes.

For each prime p , define the p -primary component

$$T_p = \{g \in T(G) : p^r g = e \text{ for some } r \geq 1\}.$$

We claim $T(G) \cong \prod_p T_p$. To see this, note that any two elements of orders p^a and q^b with $p \neq q$ interact trivially: since $\gcd(p^a, q^b) = 1$, Bezout gives integers x, y with $xp^a + yq^b = 1$, from which every element of $T(G)$ decomposes uniquely into its p -primary components.

Step 3: Decompose each T_p into cyclic groups.

Fix a prime p . We show

$$T_p \cong \mathbb{Z}_{p^{r_1}} \times \mathbb{Z}_{p^{r_2}} \times \cdots \times \mathbb{Z}_{p^{r_s}}$$

for some positive integers $r_1 \geq r_2 \geq \cdots \geq r_s$. Take $g_1 \in T_p$ of maximum order p^{r_1} , so $\langle g_1 \rangle \cong \mathbb{Z}_{p^{r_1}}$. Since a cyclic subgroup of maximum order in a p -group is always a direct summand, there exists a subgroup $H \leq T_p$ such that $T_p = \langle g_1 \rangle \times H$. Applying the same argument to H and repeating, the process terminates since G is finitely generated, yielding the desired decomposition.

Uniqueness.

The Betti number k is uniquely determined as the rank of the free group $G/T(G)$. The p -primary components T_p are uniquely determined as subgroups of $T(G)$. Within each T_p , the multiset of exponents $\{r_1, \dots, r_s\}$ is uniquely recovered from the sequence of quotients $p^i T_p / p^{i+1} T_p$. Hence the decomposition is unique up to rearrangement of factors. $\square$ $\square$

Example 2.2.3: (Classification by Primary Factors): Find all Abelian groups, up to isomorphism, of order 360.

First, observe that $360 = 2^3 \cdot 3^2 \cdot 5$. There will be no factors of $\mathbb{Z}$ since the group is finite. We look for partitions of the prime powers r_i :

- For 2^3 , the partitions are $3, 2 + 1, 1 + 1 + 1$; corresponding to $\mathbb{Z}_8, \mathbb{Z}_4 \times \mathbb{Z}_2$, and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$.
- For 3^2 , the partitions are $2, 1 + 1$; corresponding to $\mathbb{Z}_9$ and $\mathbb{Z}_3 \times \mathbb{Z}_3$.
- For 5^1 , the partition is 1 ; corresponding to $\mathbb{Z}_5$.

Combining these independent choices ($3 \times 2 \times 1 = 6$ total) yields six non-isomorphic groups:

1. $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
2. $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$
3. $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
4. $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$
5. $\mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$
6. $\mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$

Theorem 2.2.5 (Invariant Factor Version): Every finitely generated abelian group is isomorphic to a direct product of cyclic groups of the form

$$\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2} \times \mathbb{Z}_{d_3} \times \cdots \times \mathbb{Z}_{d_k} \times \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z}$$

where each $d_i \geq 2$ is an integer and d_i divides d_{i+1} for $1 \leq i \leq k-1$. Furthermore, the representation is unique.

Note:-

The numbers d_i are called the **invariant factors** (or **torsion coefficients**). Either version of the fundamental theorem can be derived from the other. Primary factors break the group down to prime-power building blocks, while invariant factors group them back up so that each cyclic component divides the next.

Example 2.2.4: (Classification by Invariant Factors): Find all Abelian groups, up to isomorphism, of order 360.

Since $360 = 2^3 \cdot 3^2 \cdot 5$, the invariant factors must multiply to 360. The possible sequences of invariant factors are:

- $d_1 = 360$ corresponding to $\mathbb{Z}_{360}$.
- $d_1 = 2, d_2 = 180$ corresponding to $\mathbb{Z}_2 \times \mathbb{Z}_{180}$.
- $d_1 = 3, d_2 = 120$ corresponding to $\mathbb{Z}_3 \times \mathbb{Z}_{120}$.
- $d_1 = 6, d_2 = 60$ corresponding to $\mathbb{Z}_6 \times \mathbb{Z}_{60}$.
- $d_1 = 2, d_2 = 2, d_3 = 90$ corresponding to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{90}$.
- $d_1 = 2, d_2 = 6, d_3 = 30$ corresponding to $\mathbb{Z}_2 \times \mathbb{Z}_6 \times \mathbb{Z}_{30}$.

These six groups correspond exactly to the six groups found using the primary factor version, confirming that both classifications yield the same set of isomorphism classes.

2.2.3 Applications

Definition 2.2.3: Decomposable and Indecomposable Groups

A group G is **decomposable** if it is isomorphic to a direct product of two proper non-trivial subgroups. Otherwise, G is **indecomposable**.

Theorem 2.2.6: The finite indecomposable Abelian groups are exactly the cyclic groups with order a power of a prime, i.e., $\mathbb{Z}_{p^r}$ for some prime p and positive integer r .

Proof:

If G is a finite indecomposable abelian group, then by the fundamental theorem, G is isomorphic to a direct product of cyclic groups of prime power order. Since G is indecomposable, it cannot be expressed as a direct product of two or more non-trivial groups, so there must be only one factor. Thus, $G \cong \mathbb{Z}_{p^r}$ for some prime p and positive integer r . Conversely, any group of the form $\mathbb{Z}_{p^r}$ is indecomposable since it cannot be expressed as a direct product of smaller non-trivial groups. $\square$

Theorem 2.2.7: If m divides the order of a finite Abelian group G , then G has a subgroup of order m .

Proof:

Let $|G| = n$ and suppose $m \mid n$. By the fundamental theorem, G is isomorphic to a direct product of cyclic groups of prime power order. We can construct a subgroup of order m by taking appropriate factors from the direct product decomposition. Since m divides n , we can express m as a product of prime powers that divide the orders of the cyclic factors in the decomposition of G . By selecting the corresponding subgroups from each factor, we can form a subgroup of G with order m . $\square$

Theorem 2.2.8: If m is a square-free positive integer (i.e., m is not divisible by the square of any prime), every Abelian group of order m is cyclic.

Proof:

Let $m = p_1 p_2 \cdots p_k$ where the p_i are distinct primes. By the fundamental theorem, any abelian group of order m is isomorphic to a direct product of cyclic groups whose orders are powers of the primes dividing m . Since each prime divides m only once, the only possible cyclic factors are $\mathbb{Z}_{p_i}$ for each prime p_i . Thus, any abelian group of order m is isomorphic to $\mathbb{Z}_{p_1} \times \mathbb{Z}_{p_2} \times \cdots \times \mathbb{Z}_{p_k}$. Since the p_i are distinct primes, this direct product is isomorphic to $\mathbb{Z}_m$, which is cyclic. Therefore, every abelian group of square-free order is cyclic. $\square$

2.3 Cosets and the Theorem of Lagrange

2.3.1 Cosets

Theorem 2.3.1: Let H be a subgroup of a group G . We can define an equivalence relation $\sim_L$ on G by:

$$a \sim_L b \iff a^{-1}b \in H$$

This relation partitions G into disjoint equivalence classes.

Proof:

We verify the properties of an equivalence relation:

- **Reflexivity:** For any $a \in G$, $a^{-1}a = e \in H$, so $a \sim_L a$.
- **Symmetry:** If $a \sim_L b$, then $a^{-1}b \in H$. Taking inverses gives $b^{-1}a = (a^{-1}b)^{-1} \in H$, so $b \sim_L a$.
- **Transitivity:** If $a \sim_L b$ and $b \sim_L c$, then $a^{-1}b \in H$ and $b^{-1}c \in H$. Multiplying these gives $a^{-1}c = (a^{-1}b)(b^{-1}c) \in H$, so $a \sim_L c$.

Thus, $\sim_L$ is an equivalence relation, and it partitions G into disjoint equivalence classes. $\square$

Definition 2.3.1: Left Coset

Let H be a subgroup of G . The subset $aH = \{ah \mid h \in H\}$ of G is the **left coset** of H containing a .

Note:

If we are using additive notation (such as in $\mathbb{Z}$), the left coset is denoted $a + H = \{a + h \mid h \in H\}$.

Example 2.3.1: Left Cosets in an Abelian Group

Let $G = \mathbb{Z}$ and $H = 3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\}$. To find the left cosets of $3\mathbb{Z}$ in $\mathbb{Z}$:

$$0 + 3\mathbb{Z} = \{\dots, -6, -3, 0, 3, 6, \dots\}$$

$$1 + 3\mathbb{Z} = \{\dots, -5, -2, 1, 4, 7, \dots\}$$

$$2 + 3\mathbb{Z} = \{\dots, -4, -1, 2, 5, 8, \dots\}$$

These three disjoint sets exhaust $\mathbb{Z}$. So there are exactly 3 distinct left cosets.

Example 2.3.2: Left Cosets in a Nonabelian Group

Consider D_4 and the subgroup $H = \langle \mu \rangle = \{\iota, \mu\}$. The left cosets are:

$$\iota H = \{\iota, \mu\}$$

$$\rho H = \{\rho, \mu\rho^3\}$$

$$\rho^2 H = \{\rho^2, \mu\rho^2\}$$

$$\rho^3 H = \{\rho^3, \mu\rho\}$$

Notice that the left coset $\rho H = \{\rho, \mu\rho^3\}$ is generally not equal to the right coset $H\rho = \{\rho, \mu\rho\}$. In abelian groups, left and right cosets are always identical.

Definition 2.3.2: Right Coset

Similarly to left cosets, we can define right cosets using the equivalence relation

$$a \sim_R b \iff ab^{-1} \in H$$

The **right coset** containing a is $Ha = \{ha \mid h \in H\}$.

2.3.2 The Theorem of Lagrange

Theorem 2.3.2: Let H be a subgroup of G . Then for any $a \in G$, the coset aH has the same cardinality as H .

Proof:

Let $|G| = n$ and $|H| = m$. Every coset of H also has m elements. Let r be the number of cells in the partition of G into left cosets of H . Then $n = rm$, so m is indeed a divisor of n . $\square$

This leads directly to one of the most famous theorems in group theory, which acts as a powerful counting tool.

Theorem 2.3.3 (Theorem of Lagrange): Let H be a subgroup of a finite group G . Then the order of H is a divisor of the order of G .

Proof:

Since there is a injection from H to any of its cosets in G , all left cosets of H have the same number of elements as H . Let $[G : H]$ be the number of distinct left cosets of H in G . Then we can express the order of G as:

$$|G| = [G : H] \cdot |H|$$

Since $[G : H]$ is a positive integer, it follows that $|H|$ divides $|G|$. $\square$

Note:-

While Lagrange's Theorem says the order of a subgroup must divide the order of the group, the **converse is not always true**. A group G of order n does not necessarily have a subgroup of order d for every divisor d of n . However, if G is finite abelian, the converse happens to hold.

Several immediate and elegant corollaries follow from Lagrange's Theorem.

Corollary 2.3.4: *Every group of prime order is cyclic.*

Proof:

Let G have prime order p , and let $a \in G$ be any element other than the identity e . The cyclic subgroup $\langle a \rangle$ has at least two elements (a and e). By Lagrange's Theorem, its order $|\langle a \rangle|$ must divide p . Since p is prime and $|\langle a \rangle| \geq 2$, we must have $|\langle a \rangle| = p$. Thus $\langle a \rangle = G$, and G is cyclic. $\square$

Note:-

Up to isomorphism, there is only one group structure of a given prime order p , which is $\mathbb{Z}_p$.

Corollary 2.3.5: *The order of an element of a finite group divides the order of the group.*

Proof:

The order of an element a is equal to the order of the cyclic subgroup $\langle a \rangle$ it generates. By Lagrange's Theorem, $|\langle a \rangle|$ divides $|G|$. $\square$

Definition 2.3.3: Index

Let H be a subgroup of a group G . The number of left (or right) cosets of H in G is the **index** of H in G , denoted by $[G : H]$.

If G is finite, the index is simply $[G : H] = \frac{|G|}{|H|}$.

Note:-

The index can also be finite even if G and H are infinite (e.g., $(\mathbb{Z} : 3\mathbb{Z}) = 3$).

Theorem 2.3.6 (Multiplicativity of Index): *Suppose H and K are subgroups of a group G such that $K \leq H \leq G$. If $[H : K]$ and $[G : H]$ are both finite, then $[G : K]$ is finite, and:*

$$[G : K] = [G : H][H : K]$$

Proof:

Let $r = [G : H]$ and $s = [H : K]$. We can write G as a disjoint union of r left cosets of H :

$$G = a_1H \cup a_2H \cup \cdots \cup a_rH$$

Each a_iH is itself a union of s left cosets of K :

$$a_iH = a_ib_{i1}K \cup a_ib_{i2}K \cup \cdots \cup a_ib_{is}K$$

Since the cosets of H are disjoint, the cosets of K obtained by this process are also disjoint. Thus, we have partitioned G into rs distinct left cosets of K , giving $[G : K] = rs = [G : H][H : K]$. $\square$

Theorem 2.3.7: Let $\phi : G \rightarrow G'$ be a group homomorphism. Then the left and right cosets of $\text{Ker}(\phi)$ are identical. Furthermore, $a, b \in G$ are in the same coset of $\text{Ker}(\phi)$ if and only if $\phi(a) = \phi(b)$.

Proof:

We first assume that a and b are in the same left cosets of $\text{Ker}(\phi)$ and show they are also in the same right cosets. Now, we've $a^{-1}b \in \text{Ker}(\phi)$. So, $\phi(a^{-1}b) = e$, the identity element. Since ϕ is a homomorphism, $\phi(a)^{-1}\phi(b) = e$, which implies that $\phi(a) = \phi(b)$. Therefore, $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = \phi(a)\phi(a)^{-1} = e$. Thus, $ab^{-1} \in \text{Ker}(\phi)$, which says that a and b are in the same right coset. Here, we showed that if a and b are in the same left coset of $\text{Ker}(\phi)$, then $\phi(a) = \phi(b)$.

Now suppose that $\phi(a) = \phi(b)$. Then $\phi(b^{-1}a) = \phi(b)^{-1}\phi(a) = e$. Thus, $b^{-1}a \in \text{Ker}(\phi)$, which implies that a and b are in the same left coset.

To complete the proof, we need to show that if a and b are in the same right coset, then they are also in the same left coset. The proof is essentially the same as above. $\square$

Corollary 2.3.8: A homomorphism $\phi : G \rightarrow G'$ is injective if and only if $\text{Ker}(\phi)$ is the trivial subgroup of G .

Proof:

We first assume that $\text{Ker}(\phi) = \{e\}$. Every coset of $\text{Ker}(\phi)$ has only one element. Suppose that $\phi(a) = \phi(b)$. Then a and b are in the same coset of $\text{Ker}(\phi)$. Thus, $a = b$.

Now suppose that ϕ is injective. Then only the identity e is mapped to the identity in G' . So, $\text{Ker}(\phi) = \{e\}$. $\square$

2.4 Plane Isometries

This section studies groups through a geometric lens, specifically by looking at mappings of the Euclidean plane that preserve distance.

2.4.1 Isometries and Their Group Structure

Definition 2.4.1: Isometry

An **isometry** of the Euclidean plane $\mathbb{R}^2$ is a permutation $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that preserves distance. That is, for any points $P, Q \in \mathbb{R}^2$, the distance between P and Q is equal to the distance between $\phi(P)$ and $\phi(Q)$.

Theorem 2.4.1: The set of all isometries of $\mathbb{R}^2$ forms a subgroup of the group of all permutations of $\mathbb{R}^2$.

Proof:

The composition of two distance-preserving maps is distance-preserving. The identity map is an isometry. The inverse of an isometry is an isometry. $\square$

Note:-

The group of isometries that carry a subset $S \subset \mathbb{R}^2$ onto itself is called the **symmetry group** of S . For a regular n -gon, this group is D_n .

2.4.2 The Four Types of Plane Isometries

It can be shown that every isometry of the plane is one of four types:

1. **Translation** (τ): Slides every point by a fixed vector.
Example: $\tau(x, y) = (x, y) + (2, -3) = (x + 2, y - 3)$
2. **Rotation** (ρ): Rotates the plane about a fixed point P .
Example: $\rho(x, y) = (-y, x)$ is a rotation through 90° counterclockwise about the origin.
3. **Reflection** (μ): Reflects the plane across a fixed line L .
Example: $\mu(x, y) = (x, -y)$ is a reflection across the x -axis.
4. **Glide Reflection** (γ): A reflection followed by a translation along the line of reflection.
Example: $\gamma(x, y) = (x + 1, -y)$ is a glide reflection across the x -axis followed by a translation of 1 unit to the right.

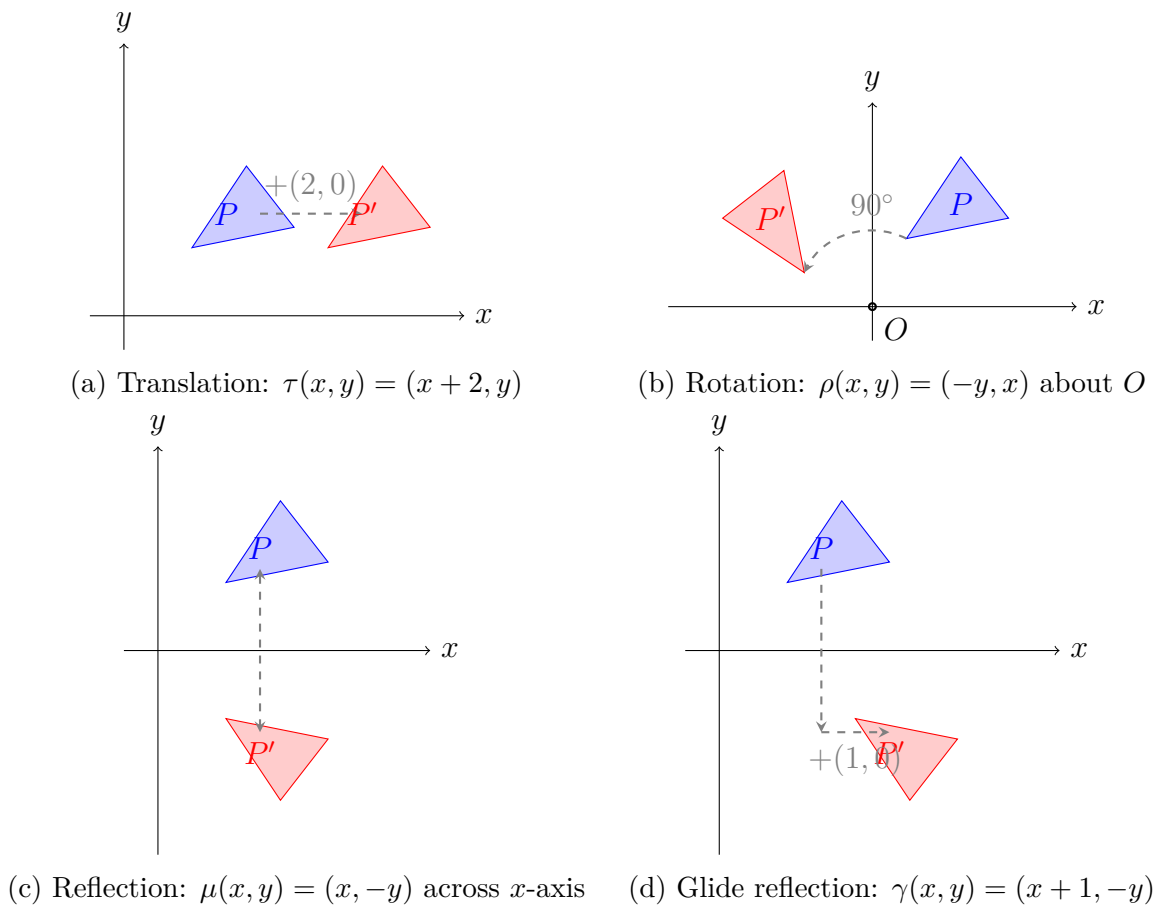


Figure 2.4.1: The four types of plane isometries

Isometries are classified as **orientation-preserving** (translations and rotations) or **orientation-reversing** (reflections and glide reflections). The orientation-preserving isometries form a subgroup of index 2 in any symmetry group that contains at least one orientation-reversing element.

2.4.3 Finite Isometry Groups

The theorem below completely classifies the structures of finite subgroups of the full isometry group.

Theorem 2.4.2 (Leonardo's Theorem): Every finite group G of isometries of the plane is isomorphic to either:

- $\mathbb{Z}_n$, the cyclic group of order $n \geq 1$, consisting of rotations.
- D_n , the dihedral group of order $2n$ ($n \geq 1$), which includes the Klein 4-group when $n = 2$.

Proof:

We outline the proof by showing that there is a point in the plane fixed by every element of finite G . Let $G = \{\phi_1, \phi_2, \dots, \phi_m\}$, and let $(x_i, y_i) = \phi_i(0, 0)$ be the images of the origin under each isometry. Consider their **center of mass** $P = (\bar{x}, \bar{y})$:

$$P = \left(\frac{x_1 + x_2 + \dots + x_m}{m}, \frac{y_1 + y_2 + \dots + y_m}{m} \right)$$

The actions of isometries in G merely permute the elements ϕ_i among themselves ($\phi_i \circ \phi_j = \phi_k$). Consequently, applying any $\phi \in G$ to the set of points $\{(x_i, y_i)\}$ simply permutes the points, leaving their center of mass P unchanged. Therefore, P is fixed by every isometry in G .

The orientation-preserving isometries in G must all be rotations about P (or the identity). Since G is finite, these rotations form a cyclic subgroup $\langle \rho \rangle \cong \mathbb{Z}_n$, where ρ is the rotation through the smallest positive angle.

If G consists solely of orientation-preserving isometries, $G = \langle \rho \rangle \cong \mathbb{Z}_n$. If G also contains an orientation-reversing isometry (a reflection across an axis passing through P), say μ , then the elements of G permute the vertices of a regular n -gon centered at P . Since the index of the rotation subgroup is 2, $G = \langle \rho \rangle \cup \mu \langle \rho \rangle$, which is isomorphic to the dihedral group D_n . $\square$

2.4.4 Infinite Isometry Groups: Friezes and Wallpaper

Beyond finite groups, we encounter groups of isometries describing patterns that repeat infinitely in one or two directions. These groups naturally arise in architecture, art, and crystallography.

Definition 2.4.2: Discrete Frieze

A **discrete frieze** consists of a pattern of finite width and height that is repeated endlessly in both directions along a baseline, forming an infinite strip (like a decorative border next to a ceiling).

All discrete frieze groups are infinite. Since the pattern repeats, there is always a minimal translation τ that slides the frieze to its next instance. This implies that every frieze group contains a subgroup generated by τ , which is isomorphic to $\mathbb{Z}$.

Example 2.4.1: Nonabelian Frieze Group D_∞

Consider an infinite sequence of integral signs spaced one unit apart:

$$\dots \int \int \int \dots$$

The symmetry group of this frieze is generated by a translation τ (one unit to the right) and a rotation ρ of 180° about the center of any integral sign. There are no reflections because of the asymmetry of the integral sign. Thus, the group is non-commutative, as $\tau\rho = \rho\tau^{-1}$. This resembles the rule in D_n , but τ has infinite order. We call this frieze

group D_∞ .

Example 2.4.2: Abelian Frieze Group $\mathbb{Z} \times \mathbb{Z}_2$

Consider an infinite string of D's:

... D D D D D ...

Its group is generated by the translation τ and a horizontal reflection μ across a line cutting through the middle of the D's. Here, the generators commute: $\tau\mu = \mu\tau$. Hence, the group is abelian and isomorphic to $\mathbb{Z} \times \mathbb{Z}_2$.

Note:-

In total, there are exactly **7 distinct types of discrete frieze groups**, classified by whether they admit horizontal reflections, vertical reflections, rotations, or non-trivial glide reflections.

If a pattern is repeated by translations along two non-parallel vector directions to fill the entire plane (like patterns on wallpaper or tiles on a floor), its symmetry group will contain a subgroup isomorphic to $\mathbb{Z} \times \mathbb{Z}$.

Note:-

It can be shown that there are exactly **17 different types of wallpaper groups** (or **plane crystallographic groups**) when classified according to the types of rotations, reflections, and glide reflections they admit.

3 Homomorphisms and Factor Groups

3.1 Factor Groups

Recall from our study of cosets that partitioning a group into left cosets of a subgroup can sometimes reveal interesting structure. We now investigate when the set of left cosets of a subgroup can itself be made into a group.

When we arrange the head on top and on the left of a group table so that the elements are grouped into left cosets of a subgroup, we may or may not obtain a coherent group table for the cosets. For the cosets to form a group, we need a crucial condition: if a_1, a_2 are in the same left coset and b_1, b_2 are in the same left coset, then a_1b_1 and a_2b_2 must be in the same left coset. When this condition holds, we say the operation on the left cosets is *induced* by the operation of G .

Example 3.1.1

Consider the group $\mathbb{Z}_6$ with subgroup $\langle 3 \rangle = \{0, 3\}$. The left cosets are:

$$0 + \langle 3 \rangle = \{0, 3\}$$

$$1 + \langle 3 \rangle = \{1, 4\}$$

$$2 + \langle 3 \rangle = \{2, 5\}$$

Observe that if $a_1, a_2 \in 1 + \langle 3 \rangle$ and $b_1, b_2 \in 2 + \langle 3 \rangle$, then $a_1 +_6 b_1$ and $a_2 +_6 b_2$ are both in $0 + \langle 3 \rangle$. This is because:

$$1 +_6 2 = 3 \in 0 + \langle 3 \rangle$$

$$1 +_6 5 = 0 \in 0 + \langle 3 \rangle$$

$$4 +_6 2 = 0 \in 0 + \langle 3 \rangle$$

$$4 +_6 5 = 3 \in 0 + \langle 3 \rangle$$

The coset operation $(aH)(bH) = (ab)H$ is well-defined, and the cosets form a group.

However, consider D_3 with subgroup $\langle \mu \rangle = \{\iota, \mu\}$. The left cosets are $\{\iota, \mu\}$ and $\{\rho, \mu\rho, \mu\rho^2\}$. When we compute the product of the coset $\{\rho\}$ times the coset $\{\rho\}$, we get the coset $\{\rho^2\}$. But if we choose different representatives from each coset, say $\mu\rho$ from the first and $\mu\rho^2$ from the second, we get $\mu\rho \cdot \mu\rho^2 = \mu(\rho\mu)\rho^2 = \mu(\mu\rho^2)\rho^2 = \mu^2\rho^4 = \rho^2$, which gives the identity coset. This inconsistency means the operation is not well-defined.

The key issue is that for coset multiplication to be well-defined, we need $gH = Hg$ for all $g \in G$. This leads us to the concept of a normal subgroup.

Definition 3.1.1: Normal Subgroup

Let H be a subgroup of G . We say that H is a **normal subgroup** of G if $gH = Hg$ for all $g \in G$. If H is a normal subgroup of G , we write $H \trianglelefteq G$.

Equivalently, H is normal if and only if $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.

Note:-

If G is abelian, then every subgroup is normal. In general, however, normality is a stronger condition than being a subgroup.

Example 3.1.2

The subgroup of even permutations $A_n \leq S_n$ is normal since A_n is the kernel of the sign homomorphism $\text{sgn} : S_n \rightarrow \{1, -1\}$.

Example 3.1.3

Let $H = \{A \in GL(n, \mathbb{R}) \mid \det(A) = 1\}$. The determinant map satisfies $\det(AB) = \det(A)\det(B)$, so $\det : GL(n, \mathbb{R}) \rightarrow \mathbb{R}^*$ is a homomorphism. Thus $H = \ker(\det)$, and $H \trianglelefteq GL(n, \mathbb{R})$. This subgroup is called the **special linear group**, denoted $SL(n, \mathbb{R})$.

The following theorem characterizes when left coset multiplication is well-defined.

Theorem 3.1.1 (Well-Defined Coset Multiplication): Let H be a subgroup of G . Then left coset multiplication is well-defined by the equation

$$(aH)(bH) = (ab)H$$

if and only if H is a normal subgroup of G .

Proof:

Suppose first that $(aH)(bH) = (ab)H$ gives a well-defined binary operation on left cosets. Let $a \in G$. We want to show $aH \subseteq Ha$. Let $x \in aH$, so $x = ah_1$ for some $h_1 \in H$. Choosing representatives $x \in aH$ and $a^{-1} \in a^{-1}H$, we have

$$(xH)(a^{-1}H) = (xa^{-1})H.$$

Choosing representatives $a \in aH$ and $a^{-1} \in a^{-1}H$, we have $(aH)(a^{-1}H) = eH = H$. By well-definedness, $xa^{-1} = h_2 \in H$ for some h_2 , so $x = h_2a \in Ha$. Thus $aH \subseteq Ha$. A symmetric argument shows $Ha \subseteq aH$.

Conversely, if H is normal, then $Hb = bH$ for all $b \in G$. Suppose we compute $(aH)(bH)$. Choosing representatives $ah_1 \in aH$ and $bh_2 \in bH$, we obtain the coset ah_1bh_2H . Since $h_1b \in Hb = bH$, we have $h_1b = bh_3$ for some $h_3 \in H$. Then

$$(ah_1)(bh_2) = a(h_1b)h_2 = a(bh_3)h_2 = (ab)(h_3h_2) \in (ab)H.$$

Thus the coset is well-defined. □

This theorem leads directly to the construction of factor groups.

Corollary 3.1.2 (Factor Group Construction): Let H be a normal subgroup of G . Then the cosets of H form a group G/H under the binary operation $(aH)(bH) = (ab)H$.

Proof:

Associativity follows from associativity in G : $(aH)[(bH)(cH)] = (aH)[(bc)H] = [a(bc)]H = [(ab)c]H = [(ab)H](cH)$. The identity element is $eH = H$ since $(aH)(eH) = (ae)H = aH$. Finally, $(aH)(a^{-1}H) = (aa^{-1})H = eH$, so $a^{-1}H$ is the inverse of aH . □

Definition 3.1.2: Factor Group (Quotient Group)

The group G/H in the preceding corollary is the **factor group** (or **quotient group**) of G by H .

Example 3.1.4

Since $\mathbb{Z}$ is abelian, $n\mathbb{Z}$ is a normal subgroup. The factor group $\mathbb{Z}/n\mathbb{Z}$ consists of the cosets

$$\mathbb{Z}/n\mathbb{Z} = \{k + n\mathbb{Z} \mid 0 \leq k < n\}.$$

This group is isomorphic to $\mathbb{Z}_n$. In fact, $1 + n\mathbb{Z}$ generates $\mathbb{Z}/n\mathbb{Z}$, so it is cyclic.

Example 3.1.5

Consider the additive group $\mathbb{R}$ and the cyclic subgroup $\langle c \rangle$ where $c > 0$. Every coset of $\langle c \rangle$ contains exactly one element x with $0 \leq x < c$. Working with these representatives, we compute addition modulo c , showing that $\mathbb{R}/\langle c \rangle$ is isomorphic to the circle group U of complex numbers of magnitude 1.

Now we connect factor groups with homomorphisms. Recall that the kernel of any homomorphism is a normal subgroup. The following theorem shows the converse: every normal subgroup is the kernel of some homomorphism.

Theorem 3.1.3 (Natural (Canonical) Homomorphism): Let H be a normal subgroup of G . Then $\gamma : G \rightarrow G/H$ defined by $\gamma(x) = xH$ is a homomorphism with kernel H .

Proof:

For $x, y \in G$, we have $\gamma(xy) = (xy)H = (xH)(yH) = \gamma(x)\gamma(y)$, so γ is a homomorphism. Since $xH = H$ if and only if $x \in H$, the kernel of γ is exactly H . $\square$

This fundamental connection between normal subgroups and homomorphisms leads to one of the most important theorems in group theory.

Theorem 3.1.4 (Fundamental Homomorphism Theorem (First Isomorphism Theorem)): Let $\phi : G \rightarrow G'$ be a group homomorphism with kernel H . Then $\phi[G]$ is a group, and $\mu : G/H \rightarrow \phi[G]$ given by $\mu(gH) = \phi(g)$ is an isomorphism. If $\gamma : G \rightarrow G/H$ is the natural homomorphism, then $\phi = \mu \circ \gamma$.

Proof:

We show μ is well-defined. If $g_1H = g_2H$, then $g_2 = g_1h$ for some $h \in H$, so $\phi(g_2) = \phi(g_1h) = \phi(g_1)\phi(h) = \phi(g_1)e' = \phi(g_1)$. Thus μ is well-defined.

To show μ is a homomorphism:

$$\mu((aH)(bH)) = \mu((ab)H) = \phi(ab) = \phi(a)\phi(b) = \mu(aH)\mu(bH).$$

Since ϕ maps onto $\phi[G]$, μ maps onto $\phi[G]$. To show μ is one-to-one, note that $\mu(aH) = e'$ if and only if $\phi(a) = e'$, which occurs if and only if $a \in H$. Thus $\ker(\mu) = \{H\}$, the trivial subgroup, so μ is injective. $\square$

Note:-

The Fundamental Homomorphism Theorem states that $\phi = \mu \circ \gamma$. This means: start with $g \in G$, map to $\gamma(g) = gH$, then map to $\mu(gH) = \phi(g)$. The map ϕ can be “factored” as the composition of the natural homomorphism followed by an isomorphism.

Example 3.1.6

Let $\phi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ be defined by $\phi(m) = m \pmod{n}$ (the remainder when m is divided by n). One can verify that ϕ is a homomorphism. The kernel is $n\mathbb{Z}$, so by the Fundamental

Homomorphism Theorem, $\mathbb{Z}/\ker(\phi) = \mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$.

Example 3.1.7

Consider the projection map $\pi_1 : \mathbb{Z}_4 \times \mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ defined by $\pi_1(x, y) = x$. This is a homomorphism onto $\mathbb{Z}_4$ with kernel $\{0\} \times \mathbb{Z}_2$. By the Fundamental Homomorphism Theorem, $(\mathbb{Z}_4 \times \mathbb{Z}_2)/(\{0\} \times \mathbb{Z}_2) \cong \mathbb{Z}_4$.

We conclude with an important characterization of normal subgroups in terms of inner automorphisms.

Definition 3.1.3: Inner Automorphism

An isomorphism $\phi : G \rightarrow G$ of a group with itself is an **automorphism** of G . The map $i_g : G \rightarrow G$ defined by $i_g(x) = gxg^{-1}$ for all $x \in G$ is the **inner automorphism** of G by g .

Theorem 3.1.5 (Alternative Characterizations of Normal Subgroups): The following are equivalent conditions for a subgroup H of G to be normal:

1. $ghg^{-1} \in H$ for all $g \in G$ and $h \in H$.
2. $gHg^{-1} = H$ for all $g \in G$.
3. There is a group homomorphism $\phi : G \rightarrow G'$ such that $\ker(\phi) = H$.
4. $gH = Hg$ for all $g \in G$.

Note:-

Condition (2) shows that normal subgroups are precisely those subgroups that are invariant under all inner automorphisms of G . A subgroup $K = gHg^{-1}$ is called a *conjugate* of H .

3.2 Factor-Group Computations and Simple Groups

Factor groups can be challenging to grasp initially, but computation strengthens understanding. The key idea is that in the factor group G/N , the normal subgroup N acts as the identity element. We can visualize collapsing each coset of N to a single point.

Example 3.2.1

Compute $\mathbb{Z}/\{0\}$. Since $\{0\}$ has only one element, every coset has only one element. Thus $\mathbb{Z}/\{0\} \cong \mathbb{Z}$; there is no collapsing at all.

Example 3.2.2

Compute $\mathbb{R}/n\mathbb{R}$ where $n > 0$. Since every real number can be written as $n(x/n)$ with $x/n \in \mathbb{R}$, we have $n\mathbb{R} = \mathbb{R}$. Thus the factor group has only one element (the identity), a trivial group.

These extremes— $G/\{e\} \cong G$ and $G/G \cong \{e\}$ —are of little importance. When $N \neq \{e\}$ is a proper normal subgroup, G/N is smaller than G and often reveals simpler structure.

Example 3.2.3

Since $|S_n| = 2|A_n|$, we have $A_n \trianglelefteq S_n$, and S_n/A_n has order 2. Renaming the coset A_n as “even” and σA_n (for any odd σ) as “odd,” the multiplication reflects the parity rules: even $\times$ even = even, even $\times$ odd = odd, odd $\times$ odd = even.

The following theorem shows that factoring out by a subgroup of a direct product often yields a simpler group.

Theorem 3.2.1 (Direct Product Factorization): Let $G = H \times K$ be the direct product of groups H and K . Then $H' = \{(h, e) \mid h \in H\}$ is a normal subgroup of G , and $G/H' \cong K$. Similarly, $K' = \{(e, k) \mid k \in K\}$ is normal and $G/K' \cong H$.

Proof:

Consider the projection $\pi_2 : H \times K \rightarrow K$ defined by $\pi_2(h, k) = k$. This is a homomorphism with kernel H' , so $H' \trianglelefteq H \times K$. Since π_2 maps onto K , the Fundamental Homomorphism Theorem gives $(H \times K)/H' \cong K$. $\square$

Example 3.2.4

Compute $(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle(0, 2)\rangle$. Here $\langle(0, 2)\rangle = \{(0, 0), (0, 2), (0, 4)\}$ has order 3. The factor group has order $|\mathbb{Z}_4 \times \mathbb{Z}_6|/3 = 24/3 = 8$.

The subgroup $\langle(0, 2)\rangle$ collapses part of the $\mathbb{Z}_6$ factor (since $(0, 2)$ generates a subgroup of order 3 in $\mathbb{Z}_6$), leaving a factor isomorphic to $\mathbb{Z}_2$. The $\mathbb{Z}_4$ factor is unchanged. Thus the result is $\mathbb{Z}_4 \times \mathbb{Z}_2$.

Example 3.2.5

Compute $(\mathbb{Z} \times \mathbb{Z})/\langle(1, 1)\rangle$. Visualize $\mathbb{Z} \times \mathbb{Z}$ as integer lattice points in the plane. The subgroup $\langle(1, 1)\rangle$ consists of points on the line $y = x$. Each coset corresponds to a line parallel to this. Choosing representatives on the x -axis gives a group isomorphic to $\mathbb{Z}$. Formally, the map $\phi : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $\phi(a, b) = a - b$ is a homomorphism onto $\mathbb{Z}$ with kernel $\langle(1, 1)\rangle$. Thus $(\mathbb{Z} \times \mathbb{Z})/\langle(1, 1)\rangle \cong \mathbb{Z}$.

Theorem 3.2.2 (Cyclic Factor Groups): If G is a cyclic group and N is a subgroup of G , then G/N is cyclic.

Proof:

Let $G = \langle a \rangle$ and let N be a subgroup. Since G is abelian, N is normal. The cyclic subgroup generated by aN in G/N is

$$\langle aN \rangle = \{(aN)^n \mid n \in \mathbb{Z}\} = \{a^n N \mid n \in \mathbb{Z}\}.$$

Since $\{a^n \mid n \in \mathbb{Z}\} = G$, we have $\{a^n N \mid n \in \mathbb{Z}\} = \{gN \mid g \in G\}$, so $G/N = \langle aN \rangle$ is cyclic. $\square$

Now we explore simple groups, which are groups with no nontrivial proper normal subgroups.

Definition 3.2.1: Simple Group

A group is **simple** if it is nontrivial and has no proper nontrivial normal subgroup.

Theorem 3.2.3 (Simplicity of Alternating Groups): The alternating group A_n is simple for $n \geq 5$.

Note:-

The proof of this theorem is involved and typically uses the action of A_n on $\{1, 2, \dots, n\}$. The key ideas are: (1) A_n is generated by 3-cycles, (2) any nontrivial normal subgroup containing a 3-cycle must be all of A_n , and (3) no nontrivial element of A_n (for $n \geq 5$) can be a product of disjoint cycles in a way that avoids generating A_n .

Note:-

Simple groups are the “building blocks” of finite group theory, analogous to prime numbers in number theory. Every finite group can be “factored” into simple groups, though the factors are not uniquely determined up to order (unlike prime factorization).

A maximal normal subgroup is one that is properly contained in G but is not properly contained in any other proper normal subgroup.

Definition 3.2.2: Maximal Normal Subgroup

A **maximal normal subgroup** of G is a normal subgroup $M \neq G$ such that there is no proper normal subgroup N of G with $M \subset N$.

Theorem 3.2.4 (Maximal Normal Subgroups and Simple Groups): M is a maximal normal subgroup of G if and only if G/M is simple.

Proof:

Let M be maximal normal. Consider the natural homomorphism $\gamma : G \rightarrow G/M$. If N is a nontrivial proper normal subgroup of G/M , then $\gamma^{-1}[N]$ is a proper normal subgroup of G properly containing M (by properties of homomorphisms and normal subgroups), contradicting maximality. Thus G/M is simple.

Conversely, if G/M is simple and N is a normal subgroup of G with $M \subset N \subset G$, then $\gamma[N]$ is a normal subgroup of G/M . Since $N \neq G$, we have $\gamma[N] \neq G/M$; since $N \neq M$, $\gamma[N] \neq \{M\}$. This contradicts simplicity of G/M . Thus no such N exists and M is maximal. $\square$

We now discuss two important normal subgroups: the center and the commutator subgroup.

Definition 3.2.3: Center of a Group

The **center** of G , denoted $Z(G)$, is

$$Z(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}.$$

Note:-

$Z(G)$ is always a normal subgroup of G , and $G/Z(G)$ is abelian if and only if $Z(G) = G$. The center of S_3 is trivial ($\{e\}$).

Definition 3.2.4: Commutator

An element of the form $aba^{-1}b^{-1}$ is a **commutator** in G .

Definition 3.2.5: Commutator Subgroup

The **commutator subgroup** C (or $[G, G]$) of G is the subgroup generated by all commutators. It is the smallest normal subgroup of G containing all commutators.

Theorem 3.2.5 (Commutator Subgroup Characterization): Let G be a group with commutator subgroup C . Then:

1. $C \trianglelefteq G$.
2. G/C is abelian.
3. If $N \trianglelefteq G$, then G/N is abelian if and only if $C \leq N$.

Proof:

To show C is normal, let $x = a_1 a_2 \cdots a_k$ be a product of commutators and let $g \in G$. Then

$$g^{-1} x g = (g^{-1} a_1 g) (g^{-1} a_2 g) \cdots (g^{-1} a_k g).$$

Each $g^{-1} a_i g$ can be expressed as a product of commutators (since $g^{-1} a b a^{-1} b^{-1} g = (g^{-1} a g) (g^{-1} b g) (g^{-1} a^{-1} g) (g^{-1} b^{-1} g)$, so $g^{-1} x g \in C$. Thus $C \trianglelefteq G$.

For (2): In G/C , we have $(aC)(bC) = abC = ab(ba^{-1}b^{-1}a)C = baC = (bC)(aC)$, so G/C is abelian.

For (3): If G/N is abelian, then $(aN)(bN) = (bN)(aN)$, so $aba^{-1}b^{-1} \in N$ for all a, b , hence $C \leq N$. Conversely, if $C \leq N$, then G/N is abelian by the same calculation. $\square$

Example 3.2.6

In S_3 , one commutator is $(1, 2, 3)(2, 3)(1, 2, 3)^{-1}(2, 3)^{-1} = (1, 3, 2)$. The subgroup generated by this is A_3 . Since $S_3/A_3 \cong \mathbb{Z}_2$ is abelian, the commutator subgroup of S_3 is A_3 .

3.3 Group Action on a Set

Group actions provide a powerful way to study groups through their effects on sets. This framework connects many previous concepts and leads to deep results like Cauchy's Theorem.

Definition 3.3.1: Group Action

Let X be a set and G a group. An **action** of G on X is a map $G \times X \rightarrow X$ (written $g \cdot x$ or simply gx) such that:

1. $ex = x$ for all $x \in X$ (where e is the identity in G).
2. $(g_1 g_2)x = g_1(g_2 x)$ for all $g_1, g_2 \in G$ and $x \in X$.

Under these conditions, X is a **G -set**.

Example 3.3.1

Let $G = GL(n, \mathbb{R})$ and let $X = \mathbb{R}^n$ (column vectors). Matrix multiplication defines an action: for $A \in GL(n, \mathbb{R})$ and $v \in \mathbb{R}^n$, $A \cdot v = Av$. The identity matrix acts as the identity, and $(AB)v = A(Bv)$.

Example 3.3.2

Let $G = D_n$ (dihedral group). Elements of D_n permute the vertices $\{0, 1, 2, \dots, n-1\}$ of a regular n -gon. For $\alpha \in D_n$ and $k \in \{0, 1, \dots, n-1\}$, the action is $\alpha \cdot k = \alpha(k)$.

Example 3.3.3

Every group G acts on itself by left multiplication: for $g \in G$ and $x \in G$, define $g \cdot x = gx$. If $H \leq G$, then G is an H -set under this action.

Example 3.3.4

Any group G acts on itself by conjugation: for $g \in G$ and $x \in G$, define $g \cdot x = gxg^{-1}$. This action satisfies $e \cdot x = x$ and $(gh) \cdot x = ghxg^{-1}h^{-1} = g(h \cdot x)h^{-1} = g \cdot (h \cdot x)$.

The following theorem shows that every group action is essentially a permutation action.

Theorem 3.3.1 (Group Actions are Permutation Actions): Let X be a G -set. For each $g \in G$, the map $\sigma_g : X \rightarrow X$ defined by $\sigma_g(x) = gx$ is a permutation of X . The map $\phi : G \rightarrow S_X$ defined by $\phi(g) = \sigma_g$ is a homomorphism.

Proof:

To show σ_g is a permutation, suppose $\sigma_g(x_1) = \sigma_g(x_2)$. Then $gx_1 = gx_2$, so $g^{-1}(gx_1) = g^{-1}(gx_2)$, giving $x_1 = x_2$. Thus σ_g is injective. For surjectivity, if $y \in X$, then $y = g(g^{-1}y) = \sigma_g(g^{-1}y)$. Hence σ_g is a permutation.

For the homomorphism property:

$$\phi(g_1g_2)(x) = \sigma_{g_1g_2}(x) = (g_1g_2)x = g_1(g_2x) = \sigma_{g_1}(\sigma_{g_2}(x)) = (\sigma_{g_1} \circ \sigma_{g_2})(x).$$

Thus $\phi(g_1g_2) = \phi(g_1)\phi(g_2)$. □

Note:-

The kernel of ϕ is $\{g \in G \mid gx = x \text{ for all } x \in X\}$, the set of elements that fix every element of X . If this kernel is trivial, the action is *faithful*.

Now we introduce two fundamental concepts associated with group actions.

Definition 3.3.2: Orbit

Let X be a G -set and let $x \in X$. The **orbit** of x under G is

$$Gx = \{gx \mid g \in G\}.$$

The set of orbits partitions X .

Definition 3.3.3: Isotropy Subgroup (Stabilizer)

Let X be a G -set and let $x \in X$. The **isotropy subgroup** (or **stabilizer**) of x is

$$G_x = \{g \in G \mid gx = x\}.$$

Theorem 3.3.2 (Isotropy Subgroup is a Subgroup): For any $x \in X$, the isotropy subgroup G_x is a subgroup of G .

Proof:

If $g_1, g_2 \in G_x$, then $(g_1g_2)x = g_1(g_2x) = g_1x = x$, so $g_1g_2 \in G_x$. The identity e satisfies $ex = x$, so $e \in G_x$. If $g \in G_x$, then $gx = x$, so multiplying by g^{-1} gives $x = g^{-1}x$, hence $g^{-1} \in G_x$. □

Theorem 3.3.3 (Orbit-Stabilizer Theorem): Let X be a G -set and let $x \in X$. Then $|Gx| = (G : G_x)$. In particular, if $|G|$ is finite, then $|Gx|$ divides $|G|$.

Proof:

Define $\psi : Gx \rightarrow \text{set of left cosets of } G_x$ by $\psi(gx) = gG_x$. This is well-defined: if $gx = hx$, then $h^{-1}gx = x$, so $h^{-1}g \in G_x$, giving $gG_x = hG_x$.

The map is injective: if $\psi(g_1x) = \psi(g_2x)$, then $g_1G_x = g_2G_x$, so $g_2 = g_1g$ for some $g \in G_x$.

Then $g_2x = g_1(gx) = g_1x$.

Surjectivity is clear: any left coset gG_x is the image of gx . Thus $|Gx| = (G : G_x)$. $\square$

Example 3.3.5

Let $G = D_4$ act on the vertices $\{0, 1, 2, 3\}$ of a square. The stabilizer of vertex 0 is $G_0 = \{\iota, \mu\}$ (the identity and the reflection fixing 0). The orbit has size $|Gx| = (G : G_x) = 8/2 = 4$, which checks out.

The orbits partition X into disjoint cells. This gives us a counting formula.

Theorem 3.3.4 (Orbit Decomposition): If X is a G -set with orbits $O_1, O_2, \dots, O_r$, then

$$|X| = \sum_{i=1}^r |O_i|.$$

Applications of group actions to finite groups yield powerful results.

Definition 3.3.4: p -Group

A **p -group** is a group in which every element has order a power of p (where p is prime).
A **p -subgroup** of G is a subgroup that is a p -group.

Theorem 3.3.5 (Cauchy's Theorem): Let G be a finite group and let p be a prime dividing $|G|$. Then G has an element of order p and hence a subgroup of order p .

Proof:

S

$\square$ etch Consider the set

$$X = \{(g_0, g_1, \dots, g_{p-1}) \mid g_i \in G \text{ and } g_0g_1 \cdots g_{p-1} = e\}.$$

There are $|G|^{p-1}$ such tuples. The cyclic group $\mathbb{Z}_p$ acts on X by cyclic permutation:

$$k \cdot (g_0, g_1, \dots, g_{p-1}) = (g_k, g_{k+1}, \dots, g_{k-1}).$$

By a counting argument (orbit decomposition modulo p), X has a nontrivial orbit. In a non-trivial orbit, all entries of the tuple are equal to some $a \in G$ with $a \neq e$ and $a^p = e$. Thus a has order p .

Theorem 3.3.6 (Center of a p -Group): If G is a finite p -group, then $Z(G) \neq \{e\}$.

Proof:

Let G act on itself by conjugation. The fixed points under this action are precisely $Z(G)$. By the orbit decomposition applied to p -groups, $|X| \equiv |X_G| \pmod{p}$. Since $|X| = |G|$ is divisible

by p and $|X_G| \geq 1$ (the identity is always fixed), $|X_G|$ must be divisible by p . Hence $Z(G)$ contains more than just the identity. $\square$

Theorem 3.3.7 (Groups of Order p^2 are Abelian): Every group of order p^2 is abelian.

Proof:

Let $|G| = p^2$ with $Z(G) \neq \{e\}$. If $Z(G) = G$, we are done. Otherwise, $|Z(G)| = p$. Since $Z(G) \trianglelefteq G$, we can form $G/Z(G)$, which has order p and is thus cyclic: $G/Z(G) = \langle aZ(G) \rangle$. Let $x, y \in G$. Then $x = a^i z_1$ and $y = a^j z_2$ for some $z_1, z_2 \in Z(G)$. Then

$$xy = a^i z_1 a^j z_2 = a^{i+j} z_1 z_2 = a^j z_2 a^i z_1 = yx.$$

Thus G is abelian, contradicting $|Z(G)| < p^2$. Hence $Z(G) = G$. $\square$

Note:-

Up to isomorphism, there are exactly two groups of order p^2 : $\mathbb{Z}_{p^2}$ and $\mathbb{Z}_p \times \mathbb{Z}_p$. For $p = 2$, these are $\mathbb{Z}_4$ and the Klein four-group V_4 .

3.4 Applications of G-Sets to Counting

Group actions provide elegant solutions to counting problems. The key insight is that when a group G acts on a set X , the orbits correspond to “essentially different” configurations when symmetries are taken into account.

Consider a problem: how many distinguishable dice can be made by marking six faces with one to six dots? There are $6! = 720$ ways to mark the faces. However, rotating the die yields the same configuration. The rotation group of a cube has 24 elements. Each orbit under this action corresponds to one distinguishable die.

Theorem 3.4.1 (Burnside's Formula): Let G be a finite group and X a finite G -set. If r is the number of orbits in X under G , then

$$r \cdot |G| = \sum_{g \in G} |X_g|,$$

where $X_g = \{x \in X \mid gx = x\}$ is the set of elements fixed by g .

Proof:

Count the set of pairs $P = \{(g, x) \mid gx = x\}$ in two ways.

Fixing g , there are $|X_g|$ pairs, so $|P| = \sum_{g \in G} |X_g|$.

Fixing x , there are $|G_x|$ pairs, so $|P| = \sum_{x \in X} |G_x|$.

By the Orbit-Stabilizer Theorem, $|G_x| = |G|/|Gx|$. Thus

$$\sum_{x \in X} |G_x| = \sum_{x \in X} \frac{|G|}{|Gx|} = |G| \sum_{x \in X} \frac{1}{|Gx|}.$$

Each orbit O contributes $\sum_{x \in O} 1/|O| = 1$ to this sum. With r orbits, we get $|P| = r|G|$. $\square$

Corollary 3.4.2: If G is a finite group and X is a finite G -set, then

$$\text{number of orbits} = \frac{1}{|G|} \sum_{g \in G} |X_g|.$$

Example 3.4.1: Counting Distinct Dice

Let X be the set of 720 possible face markings of a die. The rotation group G of a cube has $|G| = 24$. For any nontrivial rotation $g \neq e$, no marking is fixed (every face moves), so $|X_g| = 0$. For the identity, $|X_e| = 720$.

By Burnside's formula:

$$\text{distinct dice} = \frac{1}{24}(720 + 0 + 0 + \cdots + 0) = \frac{720}{24} = 30.$$

Example 3.4.2: Seating Arrangements at a Round Table

How many ways can 7 people be seated at a round table (where rotations are considered the same)?

Here $|X| = 7!$ arrangements, and $G = \mathbb{Z}_7$ acts by cyclic rotation. Only the identity fixes any arrangement (since any nontrivial rotation moves everyone), so $|X_e| = 7!$ and $|X_g| = 0$ for $g \neq e$.

$$\text{distinct arrangements} = \frac{1}{7} \cdot 7! = 6! = 720.$$

Example 3.4.3: Painting a Triangle

How many distinguishable ways can the edges of an equilateral triangle be painted using 4 colors?

The symmetry group is S_3 (order 6). There are $4^3 = 64$ total colorings.

g	$ X_g $	Reason
ι	64	All colorings fixed
$(1, 2, 3)$	4	All edges same color
$(1, 3, 2)$	4	All edges same color
$(1, 2)$	$4^2 = 16$	Edges 1,2 same; edge 3 free
$(1, 3)$	16	Edges 1,3 same; edge 2 free
$(2, 3)$	16	Edges 2,3 same; edge 1 free

$$\text{distinct colorings} = \frac{1}{6}(64 + 4 + 4 + 16 + 16 + 16) = \frac{120}{6} = 20.$$

Example 3.4.4: Painting a Triangle with Distinct Colors

Repeat the previous problem, but use all different colors (choosing 3 colors from 4).

Now $|X| = 4 \cdot 3 \cdot 2 = 24$. The identity fixes all 24 colorings, while any nontrivial permutation changes the coloring, so $|X_g| = 0$ for $g \neq e$.

$$\text{distinct colorings} = \frac{1}{6}(24 + 0 + 0 + 0 + 0 + 0) = 4.$$

Note:-

Burnside's formula reduces the problem of counting orbits to computing, for each group element, how many configurations it fixes. This often simplifies counting significantly.

Example 3.4.5: Necklaces

How many distinguishable necklaces can be made with 7 beads of different colors (from a set of n colors)?

Here $G = D_7$ (order 14) acts on the n^7 colorings by rotations and reflections. The formula becomes:

$$\text{necklaces} = \frac{1}{14} \left(n^7 + \sum_{k=0}^6 n^{\gcd(7,k)} + 7n^{(7+1)/2} \right).$$

For $n = 7$ colors and 7 beads:

$$\text{necklaces} = \frac{1}{14} (7^7 + 6 \cdot 7 + 7 \cdot 7^4) = 360.$$

Note:-

The theory of counting using group actions extends to **Pólya's Enumeration Theorem**, which provides a systematic method for computing $|X_g|$ using cycle structures of permutations. The cycle index polynomial captures this information elegantly.